

# Intuition Engine Architecture

*Last modified: 2026-07-13*

Intuition Engine is a multi-CPU fantasy computer with 6 heterogeneous CPU cores, 6 video systems, audio engines and players, a copper coprocessor, DMA blitter, and extensive I/O peripherals - all connected through a unified MachineBus. Total guest RAM is sized at boot from platform-dispatched usable-RAM detection (`/proc/meminfo` on Linux, `GlobalMemoryStatusEx` on Windows, and `hw.memsize` on Darwin) minus a per-platform reserve. Darwin RAM sizing uses a page-aligned conservative half of `hw.memsize` as the detected base before applying the per-platform reserve. Each CPU/profile sees an active visible RAM clamped to its own ceiling. Guest software discovers sizes through the `SYSINFO MMIO` pairs (`SYSINFO_TOTAL_RAM_L0/HI`, `SYSINFO_ACTIVE_RAM_L0/HI`) and `IE64 CR_RAM_SIZE_BYTES`. This document describes the system architecture with diagrams showing chips, buses, internal functional units, and data flow paths.

The diagrams below describe wired runtime behaviour. Platform availability follows the runtime dispatch path; for example, the Z80 JIT is available on amd64 builds only.

## Reading the Architecture Tables and Diagrams

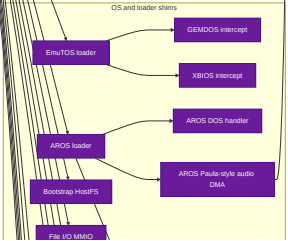
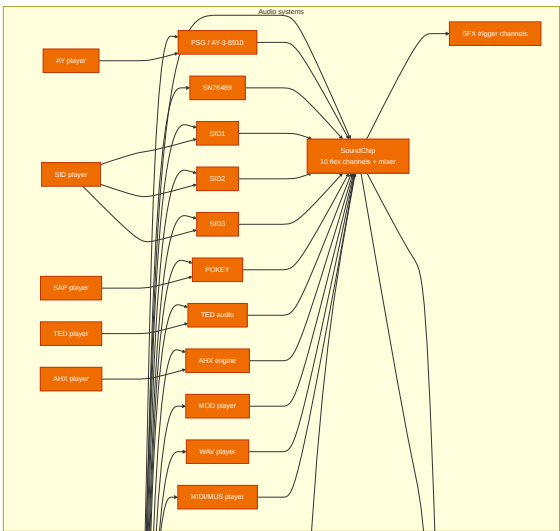
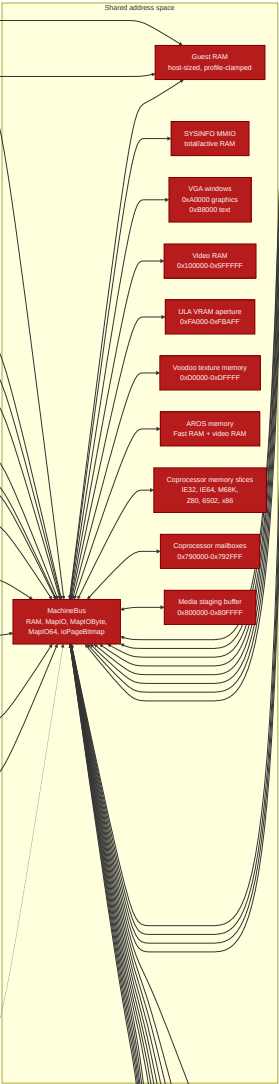
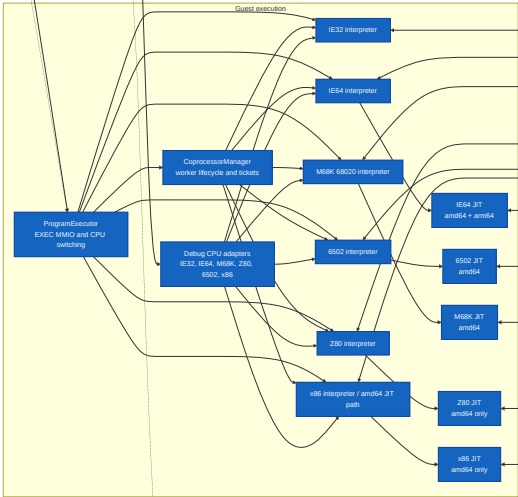
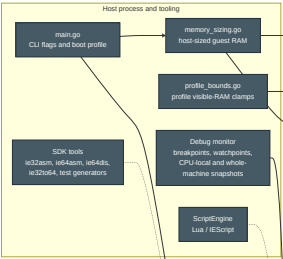
Boxes represent runtime components. Arrows represent operational paths: direct calls, bus mappings, shared-memory paths, queues, or host-service dependencies. Memory-map rows describe guest-visible address ranges and the device or subsystem that owns each range.

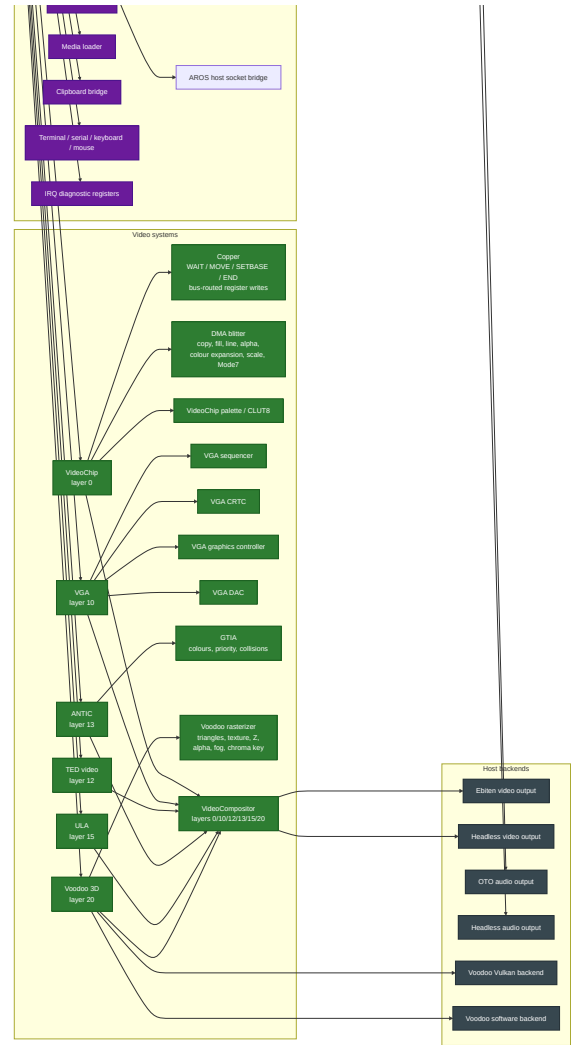
| Diagram/table item              | Meaning                                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------------|
| Address range                   | Guest-visible range plus decoded mapping, owner, and reservation/use semantics                               |
| Visible RAM size                | RAM visible to the current CPU/profile, as reported by <code>SYSINFO</code> and CPU-specific discovery paths |
| CPU or JIT entry                | CPU core or JIT backend available through the runtime dispatch path                                          |
| Audio, video, or peripheral box | Engine, player, device, or bridge that participates in the running machine                                   |
| Debugger or scripting path      | Monitor, debug adapter, or Lua binding path used for inspection or automation                                |

If a feature is platform-limited, the table or diagram names the supported platform path instead of implying that every implementation file is active in every build.

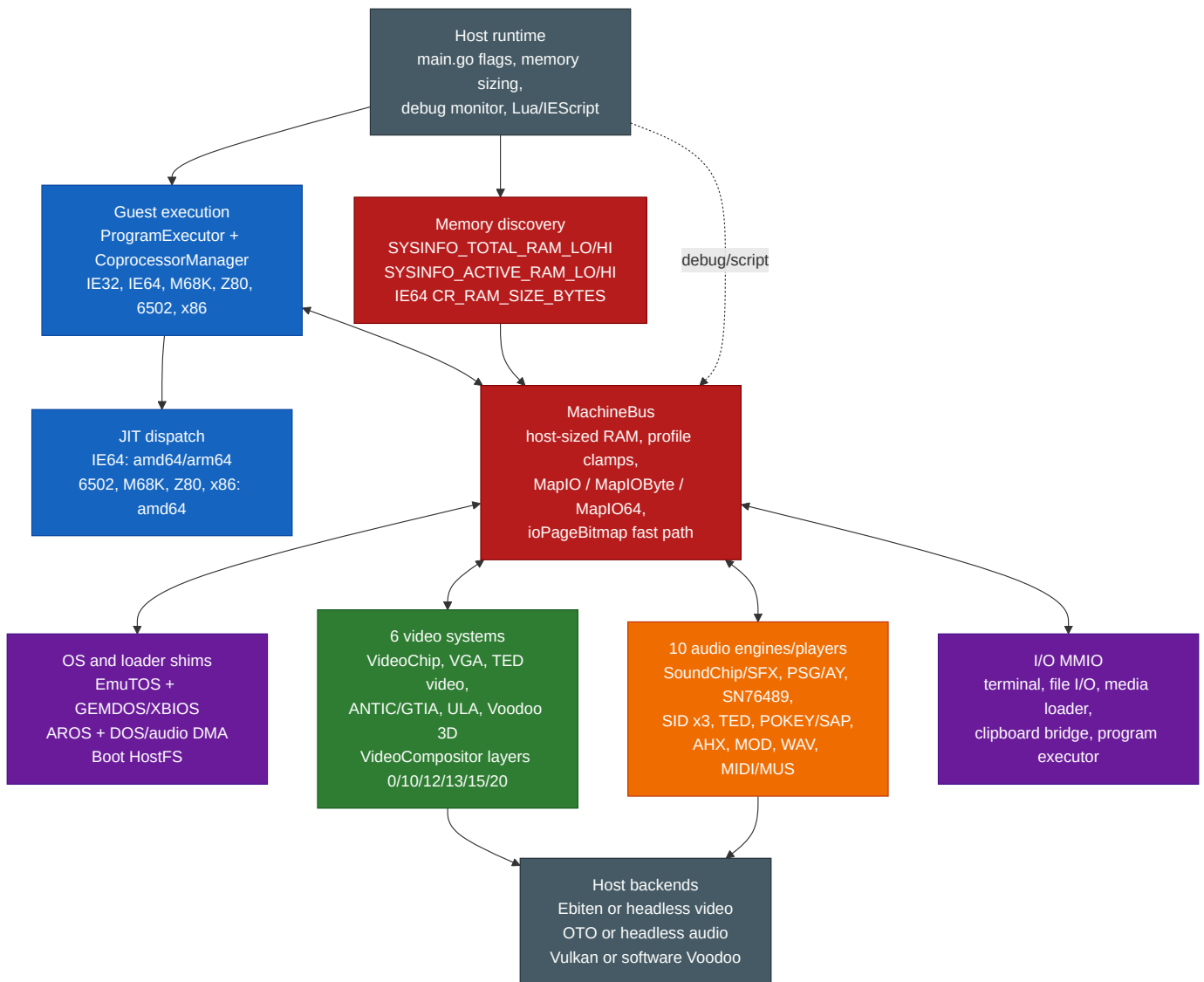
# Single Complete Architecture Diagram

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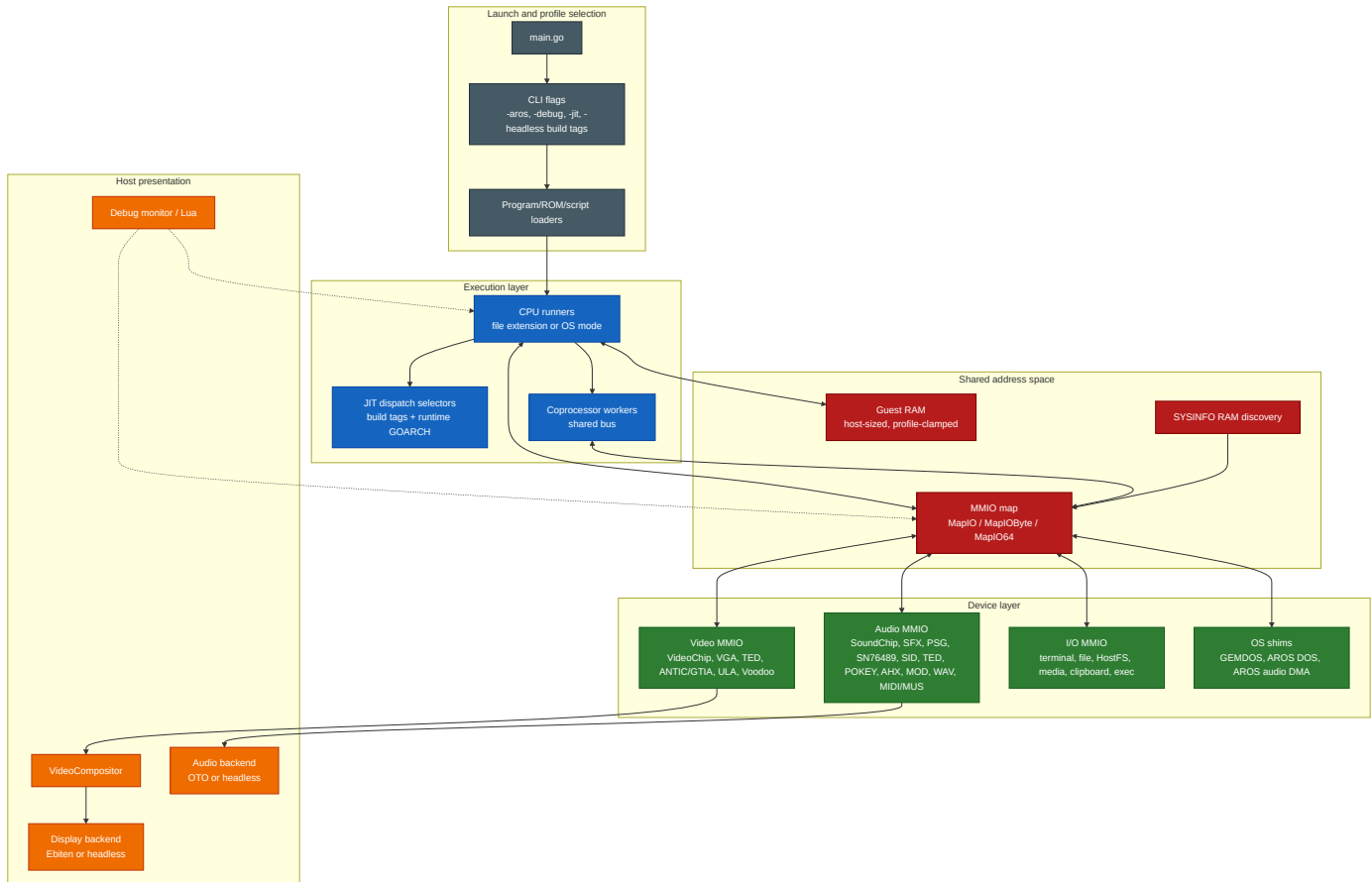




# 1. Whole-System Architecture



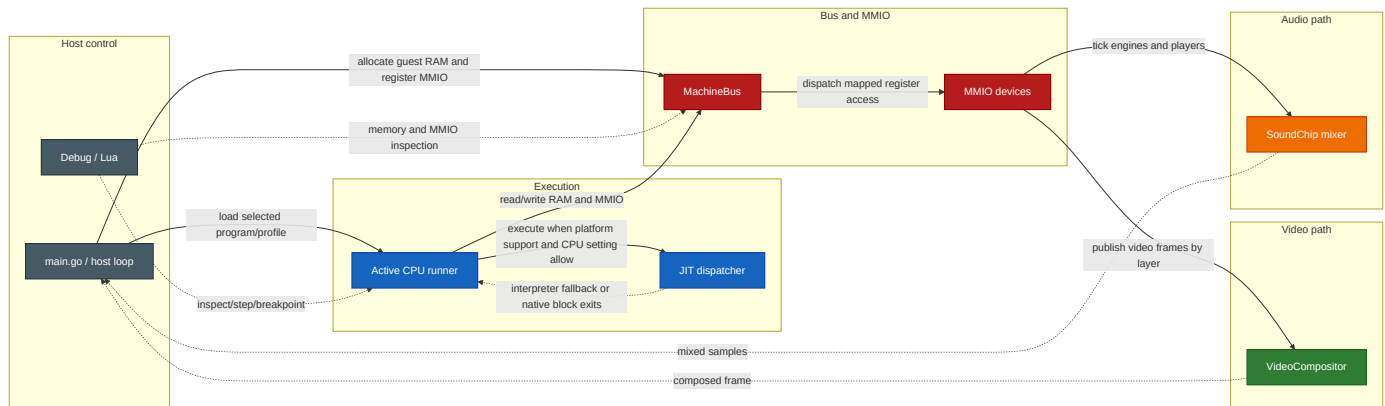
## 2. Layered System Overview



### Bus architecture notes:

- **Concurrent multi-CPU bus** - the Program Executor selects the primary CPU mode, but the Coprocessor Manager can launch additional worker CPUs that run concurrently on the same bus. Address dispatch is immutable after mapping is sealed, `ioPageBitmap` provides the fast path, and I/O callbacks protect their own mutable state. There is no central bus-arbitration model exposed to guest software.
- **No centralised interrupt controller** - each CPU has per-CPU interrupt lines (`IRQ/NMI` as `atomic.Bool`). Peripherals signal the active CPU directly.
- **MMIO dispatch** - the bus uses an `ioPageBitmap []bool` fast path (page = 256 bytes). Non-I/O pages use direct unsafe pointer access with zero dispatch overhead.

## Runtime Data and Control Flow



## Subsystem Matrix

| Subsystem         | Runtime surface                                                                  | Primary files                                                                                                                                                             | Wired registration / dispatch                                                                                                                                   |
|-------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPU cores         | IE32, IE64, M68K, Z80, 6502, x86                                                 | cpu_*.go, cpu*_runner.go                                                                                                                                                  | main.go selects runners by file extension, OS mode, or EXEC MMIO                                                                                                |
| JIT               | IE64 on amd64/arm64; 6502, M68K, Z80, x86 on amd64                               | jit_dispatch.go, jit_6502_dispatch.go, jit_m68k_dispatch.go, jit_z80_dispatch.go, jit_x86_dispatch.go                                                                     | Build tags plus runtime.GOARCH; non-supported hosts use dispatch stubs                                                                                          |
| Bus and RAM       | Host-sized guest RAM, profile clamps, MMIO, byte/64-bit handlers                 | machine_bus.go, memory_sizing.go, profile_bounds.go, sysinfo_mmio.go                                                                                                      | main.go registers devices before execution; MachineBus.SealMappings prevents late maps                                                                          |
| Machine lifecycle | Load resolution, reset quiesce, CPU/profile recreation, monitor/runtime rewiring | machine_lifecycle.go, main.go                                                                                                                                             | main.go owns concrete devices; Machine applies reset/load orchestration through injected dependencies and profile targets                                       |
| Video             | VideoChip, VGA, TED video, ANTIC/GTIA, ULA, Voodoo                               | video_chip.go, video_vga.go, video_ted.go, video_antic.go, video_ula.go, video_voodoo.go                                                                                  | main.go maps each register/VRAM block and registers compositor layers 0/10/12/13/15/20                                                                          |
| Audio             | SoundChip/SFX, PSG/AY, SN76489, SID x3, TED, POKEY/SAP, AHX, MOD, WAV, MIDI/MUS  | audio_chip.go, sfx_trigger.go, psg_engine.go, sn76489_chip.go, sid_engine.go, ted_engine.go, pokey_engine.go, ahx_player.go, mod_player.go, wav_player.go, midi_player.go | main.go maps chip/player MMIO and registers sample tickers/mixers into SoundChip; register-mapped players share PlayerControlState for staged playback requests |
| OS integration    | EmuTOS, AROS, GEMDOS/XBIOS, AROS DOS, Paula-style DMA                            | emutos_loader.go, aros_loader.go, gemdos_intercept.go, aros_dos_intercept.go, aros_audio_dma.go                                                                           | OS modes install intercept MMIO and loader state during boot/reset                                                                                              |
| Tooling           | Assemblers, disassembler, transpiler, generators                                 | assembler/, internal/ie64meta/, cmd/gen_ie64_opmeta/, cmd/ie32to64/, cmd/gen_m68k_cputest/, cmd/gen_interp6502/                                                           | Makefile builds SDK tools into sdk/bin/; IE64 opcode constants and name tables are generated from metadata                                                      |

## Internal Ownership Notes

- **IE64 opcode metadata** - `internal/ie64meta/table.go` is the source of truth for generated IE64 opcode constants and mnemonic/name tables used by the runtime CPU, monitor disassembler, standalone assembler, standalone disassembler, and internal assembler package. Build commands for `ie64asm` and `ie64dis` build `./assembler` with the relevant tag so generated files are compiled with the tagged tool entry point.
- **Machine lifecycle** - Machine in `machine_lifecycle.go` owns reset/load sequencing through dependency fields and profile targets. It preserves the existing guest-visible reset contract while keeping quiesce, failed-load rollback, CPU recreation, profile loading, and monitor/runtime rewiring out of the host event loop. A hard reset restages the configured coprocessor service after coprocessor reset and before CPU restart, so the service name pointer and worker-start path remain available across reset.
- **Audio playback control** - register-mapped music players use `PlayerControlState` for staged pointer/length registers, optional high pointer, bus reads, busy/error/loop state, subsong selection, and async generation invalidation. This covers SID, PSG/SN/SNDH, TED, POKEY/SAP, AHX, MOD, WAV, and MIDI/MUS without changing their public MMIO contracts.
- **Video scheduling** - `VideoScheduler` centralises ticker ownership for the compositor and migrated video render-loop shims. Tests can drive it manually; migrated engine files do not own `time.NewTicker` directly.
- **Voodoo software wrappers** - headless and novulkan builds share `softwareVoodooBackend` forwarding. The tagged Voodoo files keep feature registration and constructor selection only, so their `VoodooBackend` method sets stay identical.
- **IE64 MMU walk sharing** - CPU translation and bootstrap hostfs guest-pointer translation share the page-table walk result (`MMUSharedWalkResult`). CPU policy applies trap causes and A/D updates; hostfs policy requires `PTE_U` and leaves A/D bits untouched.
- **Performance accounting** - `IE_PERF_ACCT=1` enables per-CPU JIT/interpreter time, instruction, subsystem, and deopt counters; disabled accounting avoids atomic hot-path updates. Deopt reasons are unsupported, helper, mmio, smc, interrupt, cache\_pressure, and debug. These counters are diagnostics only and do not change guest-visible execution.

## Platform JIT Matrix

The host-side JIT support is intentionally asymmetric and follows the dispatch files, not emitter-file presence. Release amd64 builds target x86-64-v3 (`G0AMD64=v3`, the Makefile default) for the best Go-compiler codegen (AVX2, BMI1/2, FMA), but this is a recommendation, not a hard requirement: the JIT only *requires* SSE4.1, which it emits unconditionally for IE64 FINT (ROUNDSS). The amd64 backend detects this at runtime via `CPUID`; on a host without SSE4.1 `initJIT` (via `checkJITHostFeatures`) declines to enable the JIT and the CPU falls back to the interpreter rather than aborting, so plain `go run . / go build` at the default `G0AMD64=v1` works on any x86-64 CPU (older hosts run interpreted). AVX2/BMI2/LZCNT paths are separately runtime-gated via `x86Host` and are never emitted unless the host supports them.

| Host platform     | JIT-enabled guest cores      | Dispatch files                                                                                    |
|-------------------|------------------------------|---------------------------------------------------------------------------------------------------|
| Linux amd64       | IE64, 6502, M68K, Z80, x86   | <code>jit_dispatch.go</code> , amd64 per-core dispatch files                                      |
| Linux arm64       | IE64                         | <code>jit_dispatch.go</code> ; Z80 dispatch compiles but keeps <code>z80JitAvailable</code> false |
| Windows amd64     | IE64, 6502, M68K, Z80, x86   | amd64 per-core dispatch files                                                                     |
| Windows arm64     | IE64                         | IE64 dispatch only; per-core non-IE64 stubs                                                       |
| macOS amd64       | IE64, 6502, M68K, Z80, x86   | amd64 per-core dispatch files                                                                     |
| macOS arm64       | IE64                         | IE64 dispatch plus Darwin arm64 JIT write-protect helpers                                         |
| Browser (js/wasm) | IE64 (wasm bytecode backend) | <code>jit_exec_wasm.go</code> , <code>jit_wasm_runtime.go</code>                                  |



On macOS amd64, the JIT reuses the shared x86-64 host backends. On macOS arm64, executable memory uses the native MAP\_JIT model with thread-pinned write protection toggles, and non-IE64 guest cores remain interpreter-only on arm64 hosts.

On js/wasm the native backends are absent (the browser gives Go no executable memory), and IE64 has a wasm bytecode backend instead; see "IE64 wasm JIT backend" below. The other guest cores interpret in the browser.

## IE64 wasm JIT backend (js/wasm)

The browser build cannot emit machine code, but it can emit WebAssembly and let the browser's engine compile it. The wasm backend keeps the native JIT's shape and reuses its shared front end:

- **Scanner and context.** Blocks come from the same `scanBlock` and operate on the same `JITContext` through the `jitCtxOff*` field offsets. `GOARCH=wasm` is a 64-bit Go target (8-byte `uintptr`), so the layout is identical; a layout guard test (`jit_wasm_ctx_layout_test.go`) pins this on both native and js runs.
- **Encoder and translator.** `jit_wasm_encoder.go` writes wasm binaries; `jit_wasm_ie64_emit.go` translates a scanned block into one wasm function with signature `(ctxPtr i32) -> ()`. Both are untagged pure Go, so the differential test suite (`jit_wasm_ie64_diff_test.go`) executes generated blocks under wazero on native hosts and compares registers, PC, retired counts and guest RAM against the interpreter, instruction class by instruction class. Interpreter semantics are the reference wherever the amd64 emitter differs (Q-size ALU immediates zero-extend).
- **Shared memory.** Generated modules import the machine's own linear memory (`env.mem`; the hosting page exposes `instance.exports.mem` as `__goMem`), so blocks read and write the register file, guest RAM and the context in place at the same addresses the native backends use.
- **Exit protocol.** Blocks write `RetPC/RetCount` and return. MMIO accesses, out-of-window addresses and stack faults exit through the same `HELPER_*` protocol as the native backends; the Go-side dispatcher (`jit_wasm_runtime.go`) services the helper via the interpreter's memory and stack primitives and resumes. Generated code never calls back into Go.
- **Instruction coverage.** An explicit allowlist (`wasmSupportedOpcode`) delimits the backend: the MMU-off integer core (data movement, ALU, load/store, conditional branches, `BRA/JMP/HALT`, `JSR/RTS/PUSH/POP/JSR_IND`) plus FP64 (`DMOV/DLOAD/DSTORE`, arithmetic, `DABS/DNEG/DSQRT/DINT`, `DCMP` and the integer converts, with FPSR condition codes and sticky exception flags replicated bit-exactly). A hot block is compiled up to its longest supported prefix; anything else, including the FP64 transcendentals, stays on the interpreter.
- **MMU gate.** While `cpu.mmuEnabled` is true the runtime neither enqueues compiles nor enters installed blocks; both halves of the gate are tested under node.
- **Self-modifying code.** Generated STOREs (including `PUSH/JSR` return address writes) probe the code-page bitmap exactly like the native emitters; a dirty store commits, publishes its exact address range and ends the block. The runtime resolves the range against a page-to-blocks index and drops only the blocks whose compiled bytes were actually written. A probe hit that overlaps no block is a false share (data sharing a 256-byte page with compiled code, which the EhBASIC image contains) and drops nothing; treating false shares as full flushes put RUN AOT into a permanent recompile storm.
- **Tiering.** The dispatcher (`jit_exec_wasm.go`) interprets through `StepOne` and counts visits to block-start PCs; at the hot threshold a compile is submitted through `WebAssembly.instantiate`, which is asynchronous. Promises resolve during the cooperative yield, so block installation costs the machine nothing. The backend is on by default; `IE64_WASM_JIT=0` (mapped from `/demo/?jit=0`) disables it, and it also stays off when the hosting page does not expose `__goMem`.

Dispatch between compiled blocks stays inside wasm: a driver module (`wasmBuildDriverModule`) reads the next PC from the context, looks it up in a direct-mapped PC-to-slot cache in linear memory and `call_indirects` through a shared funcref table until the chain budget runs out, the lookup misses, or a block reports a helper exit or an SMC invalidation. The measured basis (node, V8): roughly 15 ns per in-wasm indirect call against roughly 700 ns per JS-boundary Invoke,

which is why per-block Go round trips are reserved for helper exits. On the node hot-loop benchmark (TestWasmJIT\_Node\_EndToEndParity, 2 million iterations) the chained JIT runs 5.3 times faster than the interpreter.

## IE64 JIT 64-bit Execution Model

The IE64 JIT (amd64 and arm64) executes code, accesses data, and operates the stack at any uint64 virtual or physical address, matching the interpreter. There is no uint32 truncation on the PC, data addresses, stack pointer, branch targets, or chain targets. The one stack exception is the amd64 non-MMU fused-leaf path (documented below): it raw-indexes [MemBase+SP] and so assumes the SP is in the low window.

SEI64 and CLI64 are emitted as per-instruction interpreter bails (not NOPs) so they mutate interruptEnabled under the JIT. External device interrupts are delivered through a record-only sink and a pending mask polled at instruction and block boundaries by both the interpreter and the JIT dispatcher; see the "External Interrupt Delivery" section of this manual for the full model.

- **Low memory window:** each guest sees a contiguous low RAM window backed by the dense `cpu.memory[]` slice. All full-machine modes, including the IE64 family, size the window at the full 32-bit range (`busMemMaxBytes`) so a flat 32-bit program launched from the BASIC REPL can occupy the same address space it would get from a direct boot; on mmap-backed hosts the window is reserved lazily and resident memory grows only with pages actually touched, while `busMemBootClamp` keeps non-mmap hosts at a heap-safe size. The actual `len(bus.memory)` is `min(autodetected TotalGuestRAM, busMemCap)` (`main.go`), so on a small host the window can be smaller than the cap. Addresses above the window resolve through the bus's 64-bit physical path, backed by the Backing interface; the split derives from `len(bus.memory)`, so the two regions are complementary by construction. IE64 BASIC keeps its whole interpreter arena (programme text, variables, strings, stack) inside the dense 32-bit window so every pointer stays 32-bit clean; the arena simply grows to fill that window, up to just below the resident stack. The SYSINFO low-window register pair (0xF2414/0xF2418) reports the dense window size so the stack, heap top and AOT arena are all bounded by it rather than by the larger backed total. Production IE64 boots on Linux/darwin bind high RAM via `MmapBacking`; `SparseBacking` is the test implementation. `SparseBacking` uses a sparse two-level atomic page table: missing pages read as zero, writes allocate leaves and pages with compare-and-swap, same-page word operations read or write the page slice directly, and reset publishes a fresh empty table. Hot RAM accesses load each I/O bitmap snapshot once per word or long access. After mappings are sealed, the bus reuses a quiescence-published snapshot pointer on the hot path; debug access also has a quiescence-updated plain active mirror with the service atomic retained as a backstop. M68K JIT invalidation is called through the bus's cached invalidator when one is registered. High physical addresses are supported for code fetch and for data / stack *operands* (the latter via the helper exit, including a high SP), with one caveat below.
- **High-PC stack-op caveat:** a block  *fetched from high physical backing that itself contains a stack op* (PUSH/POP/JSR/RTS/JSR\_IND) is not JIT-compiled -- `ExecuteJIT` takes the `highPhys && containsStackOp(instrs)` path and runs the block one instruction at a time through `interpretOne()` (which routes the stack through the bus). This is pinned by `TestJIT_HighPC_StackOpBlock_BailsToInterpreter`. Stack ops in blocks fetched from the low window are JIT-compiled and, for a high SP, route through the helper exit -- **except** the fused-leaf path below.
- **Fused-leaf high-SP exception:** on amd64 with the MMU off, `scanBlock` may fuse a JSR64 to a small leaf subroutine (`ie64FusedJSRLeafCall` / `ie64FusedRTSLeafReturn`, gated by `ie64ScanJSRLeafFusionEnabled` -- amd64 only). The fused emit (`jit_emit_amd64.go`) handles the call/return push/pop as **raw** [MemBase+SP] traffic *before* the normal `emitJSR_AMD64/emitRTS_AMD64` high-SP guards, so it does **not** take the helper exit. This is safe only because the marker is suppressed whenever `mmuBail` is set, but `mmuBail` for fused leaves is set by `compileBlockMMU` (MMU mode) only -- in non-MMU mode a fused leaf with an SP outside the low window would read/write past `cpu.memory[]`. Non-MMU guests are therefore expected to keep the stack in the low window; high-SP correctness via the helper exit holds for unfused stack ops, not for fused leaves.

- **Direct fast path vs. helper exit:** native code uses a direct `[memBase+addr]` access only when the MMU is off **and** the address is in the low window (size-aware bound: `addr <= MemSize - accessBytes`). Otherwise it takes the JITContext-mediated *helper exit*.
- **Helper exit protocol:** native code cannot safely call back into Go (it runs on g0 via `asmgocall`), so on a high/MMU access it writes a structured request to the JITContext (`NeedHelper`, `HelperAddr`, `HelperSize`, `HelperVal`, `HelperRd`, `HelperPC`, `LiveSP`), flushes the live SP and the faulting instruction's PC, and returns through the epilogue. `ExecuteJIT` dispatches the request (`handleJITHelper`) and re-enters the JIT. Helper ops cover `LOAD/STORE`, `FLOAD/FSTORE`, `DLOAD/DSTORE`, `PUSH/POP`, and the control-flow `JSR/RTS/JSR_IND`.
- **Helper resume:** when a helper retires exactly one instruction and the CPU state still matches the emitted continuation, `ExecuteJIT` may resume directly at that native continuation. IE64 helper resume is enabled by default and can be disabled with `IE64_JIT_RESUME=0`, `false`, `off`, or `no`. IE64 helper resume is cancelled by timer delivery, debug breakpoints, pending invalidation, PC changes, MMU mode changes, or PTBR changes.
- **Interpreter parity:** helpers use `cpu.loadMem/storeMem` for data and `cpu.mmuStackWrite/mmuStackRead` for the stack -- the same paths as the interpreter. `JSR` sets PC to the call target, `RTS` sets PC to the popped return address, and `JSR_IND` sets PC to `rs + sext(imm32)` -- the displacement is sign-extended, so a negative imm32 (bit 31 set) subtracts (with `rs == R31` based on the decremented SP). The FP load/store side effects differ: `FLOAD` and `DLOAD` set the destination FP register **and** the FP condition codes; `FSTORE` and `DSTORE` read the FP register and write memory, leaving FP registers and condition codes unchanged -- matching the interpreter in each case.
- **MMU rule:** when the MMU is enabled, non-atomic data/stack ops take the helper exit (virtual-to-physical mapping is not a direct offset). The dispatcher refreshes `ctx.MMUEnabled` before every `callNative` so the native guards are never stale.
- **JIT MMU micro-TLB:** IE64 JIT MMU helpers use a four-entry read/write micro-TLB for translated low-window RAM pages. Helper dispatch fills the cache only after a successful translation to dense RAM below `IO_REGION_START`; high physical backing and MMIO are not inserted. Native probes include the access type and CPU privilege/SUA/SKAC/SKEF mode bits in the key. `TLBFLUSH` clears the IE64 JIT micro-TLB and `TLBINVAL` invalidates matching micro-TLB VPN entries. Dispatch refreshes the mode prefixes before every native call.
- **Fault contract:** the handler sets `cpu.PC = HelperPC` before the operation, so `trapFault` records the correct `faultPC`; faulting instructions are not counted and re-execute after the trap. Pre-decrement ops (`PUSH/JSR/JSR_IND`) roll the SP back on a trapping fault.
- **Atomics carve-out:** atomics (`CAS/XCHG/FAA/FAND/FOR/FXOR`) do not use the helper-exit protocol. The native emitters run sequentially-consistent host-atomic fast paths only for aligned, non-MMU, low-window RAM addresses; MMU-on, high-address, MMIO, or unaligned cases bail to the interpreter so the canonical `atomicRMW64` trap and bus semantics apply. `compileBlockMMU` also sets `mmuBail` on fused `JSR/RTS` leaf markers, so under MMU the `OP_JS64/OP_RTS64` emit takes the `mmuBail` branch to `emitBailToInterpreter` (a whole-instruction interpreter bail with its own fault/accounting path) -- neither the raw fused fast path nor the guarded helper exit runs. In non-MMU mode `mmuBail` is unset, so the raw fused fast path is used (see the fused-leaf high-SP exception above).
- **Range-scoped self-modifying code:** IE64 self-modifying-code tracking uses 256-byte guest code pages and physical code-page tracking for MMU-compiled blocks. The 256-byte page mark is a prefilter: helper-side writes and native-store dispatcher exits with a known nonzero virtual range validate that range against `JITBlockCoveredRanges` before mutating the cache. Non-MMU checks scan only the non-MMU cache; MMU checks are scoped to the current PTBR, so a same-VA block in another address space does not cancel helper resume, record an SMC deopt, increment invalidation stats, or evict cached code. When a compiled store overlaps cached code, the dispatcher invalidates only the changed guest range; MMU stores that overlap compiled physical backing through an alias still fall back to whole-cache invalidation. x86 self-modifying-code tracking uses 256-byte code pages and range invalidation.

## Build Profiles and Observable Runtime

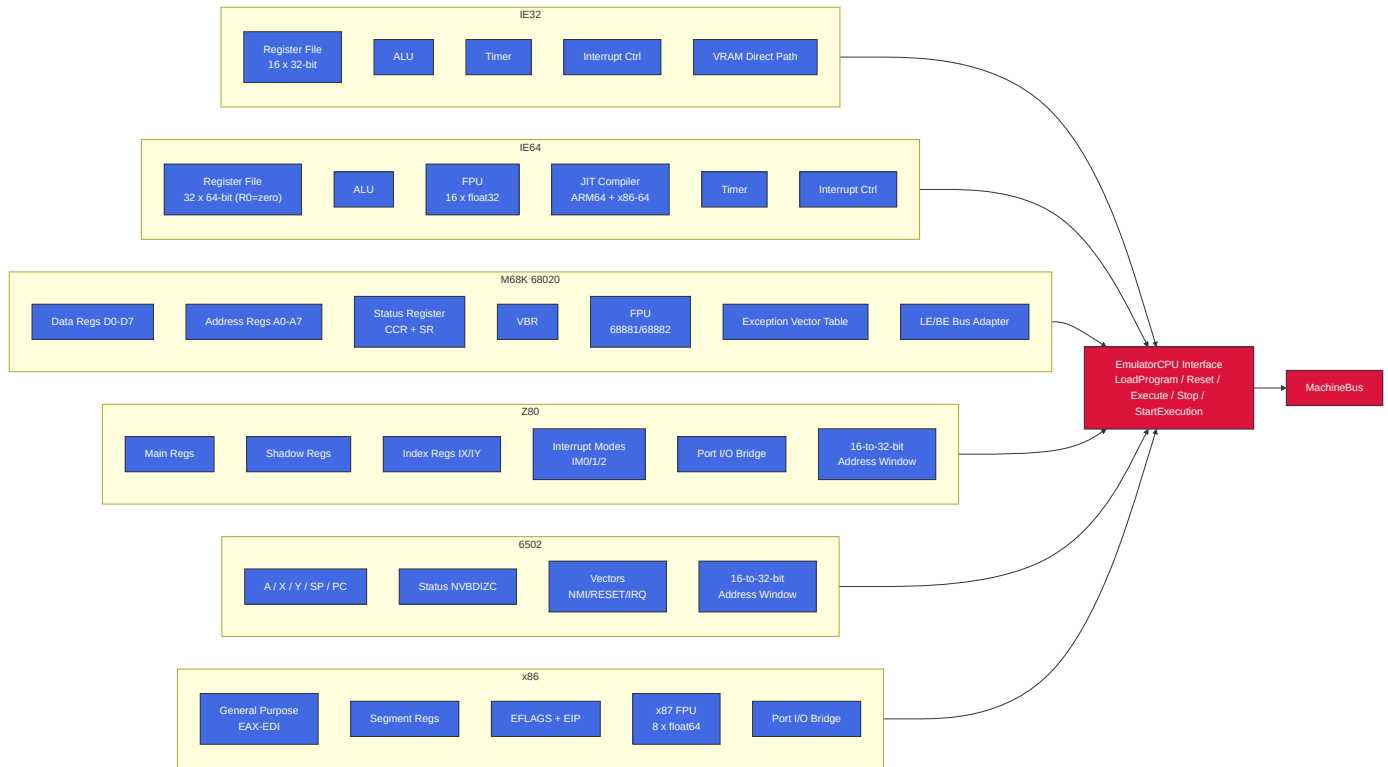
Build tags change host backends, not the guest-visible ISA contract. The main guest bus, CPU cores, MMIO register addresses, assemblers, and script binding names remain the reference surface unless a specific backend is absent from the build. On amd64, release profiles target x86-64-v3 for codegen quality; lower GOAMD64 levels still build and run, with SSE4.1 as the enforced runtime floor.

| Profile                | Build tags / knobs                       | Runtime effect visible to users                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default VM             | no special tag                           | Ebiten display, Oto audio, Vulkan-backed Voodoo where available, and normal host integrations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| novulkan               | -tags novulkan                           | Voodoo uses the software backend and does not require the Vulkan SDK. Guest Voodoo registers remain mapped.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| headless               | -tags headless                           | Display, audio backend, overlay, clipboard, and GUI integrations use stubs suitable for CI. CPU, bus, MMIO, scripting, and most device state paths still compile for tests.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| headless-novulkan      | CGO_ENABLED=0 -tags "novulkan headless"  | Pure-Go portable VM build with headless stubs and software Voodoo path.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Browser<br>(make wasm) | GOOS=js<br>GOARCH=wasm -tags embed_basic | IE64 uses a WebAssembly bytecode JIT for supported MMU-off integer and FP64 blocks, with interpreter fallback for unsupported instructions; the other CPU cores interpret.<br>IE64_WASM_JIT=0 disables the browser JIT. Ebiten renders to a WebGL canvas, Oto uses WebAudio, Vulkan is excluded, and guest RAM is a fixed 256 MiB heap backing. FileIO and Bootstrap HostFS use an in-memory volume seeded from web assets, with file contents fetched lazily on first read. CPU execution yields cooperatively so browser events, asynchronous compilation, video, and audio continue on the single WebAssembly thread. |

Headless stubs should be treated as backend substitutes, not as a different machine model. A test can still write video or audio MMIO and inspect guest state, but there is no host window, host audio device, or GUI overlay to observe the result directly.

Browser builds use an IE64 WebAssembly bytecode JIT for supported MMU-off integer and FP64 blocks, with interpreter fallback and IE64\_WASM\_JIT=0 as the runtime disable switch. Browser FileIO and Bootstrap HostFS use in-memory volumes seeded from web assets, with file contents fetched lazily on first read.

### 3. CPU Subsystem



#### CPU Timers (IE32 / IE64)

Both IE32 and IE64 use CPU-internal countdown timers backed by atomic fields on the CPU structs.

- IE32 increments an internal cycle counter once per decoded instruction step, after fetch and operand resolution and before the decoded instruction body executes; when that counter reaches `SAMPLE_RATE`, it resets the counter and decrements the timer count by one.
- IE64 decrements the timer count once per decoded instruction step, after fetch/decode and before the decoded instruction body executes. When the IE64 timer is armed, JIT dispatch uses the CPU single-instruction path so an interrupt can be delivered before the body of the instruction whose decoded step expires the timer.
- On expiry, it sets timer state to expired and can trigger a CPU interrupt.
- If still enabled after interrupt handling, the count auto-reloads from the configured period.

There is no stable bus/MMIO timer control ABI at present. Legacy include symbols such as `TIMER_CTRL/TIMER_COUNT/TIMER_RELOAD` are retained for compatibility but are deprecated.

#### CPU Selection by File Extension

| Extension                 | CPU  | Address Space                               | Notes                                             |
|---------------------------|------|---------------------------------------------|---------------------------------------------------|
| <code>.iex / .ie32</code> | IE32 | 32-bit flat (clamped to active visible RAM) | Native RISC, 8-byte fixed instructions            |
| <code>.ie64</code>        | IE64 | 64-bit (sees full active visible RAM)       | 64-bit RISC, R0=zero, JIT on ARM64 + x86-64       |
| <code>.ie68</code>        | M68K | 32-bit flat (clamped to active visible RAM) | Bare 68020, big-endian with LE bus adapter        |
| <code>.ie65</code>        | 6502 | 16-bit + bank windows                       | Bank windows reach the banked-CPU visible ceiling |
| <code>.ie80</code>        | Z80  | 16-bit + bank windows + port I/O bridge     | Bank windows reach the banked-CPU visible ceiling |

| Extension   | CPU    | Address Space                                                            | Notes                                                                     |
|-------------|--------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|
| .ie86       | x86    | 32-bit flat (clamped to active visible RAM) + compatibility bank windows | Ordinary addressing is flat; bank/VRAM windows are compatibility overlays |
| .tos / .img | M68K   | EmuTOS profile (EmuTOS_PROFILE_TOP)                                      | EmuTOS boot with GEMDOS intercept                                         |
| .ies        | Script | N/A                                                                      | Lua scripting engine (IE Script)                                          |

Bare .ie68 uses the active-visible RAM ceiling; EmuTOS and AROS M68K loader modes use profile bounds.

AROS boot is selected by CLI mode (-aros, optional -aros-image), not file extension.

## Z80 Port I/O Bridge

Z80 IN/OUT instructions are translated to bus MMIO accesses by the Z80BusAdapter:

| Z80 Port                      | Chip      | Bus Target                  | Protocol                                                                                           |
|-------------------------------|-----------|-----------------------------|----------------------------------------------------------------------------------------------------|
| \$A0-\$AD                     | VGA       | 0xF1000                     | Direct register map (MODE, STATUS, CTRL, SEQ, CRTIC, GC, DAC, DAC read index, DAC mask, VRAM bank) |
| \$B0-\$B7                     | Voodoo 3D | 0xF8000                     | Addr lo/hi + 4 data bytes (write to \$B5 triggers 32-bit bus write)                                |
| \$D0/\$D1                     | POKEY     | 0xF0D00                     | Register select / data                                                                             |
| \$D4/\$D5                     | ANTIC     | 0xF2100                     | Register select / data (x4 stride)                                                                 |
| \$D6/\$D7                     | GTIA      | 0xF2140                     | Register select / data (x4 stride, collision regs through 0xF21F8)                                 |
| \$E0/\$E1                     | SID       | 0xF0E00/0xF0E30/0xF0E50     | Register select / data; select bits 5-6 choose SID1/SID2/SID3                                      |
| \$E4/\$E5                     | SN76489   | 0xF0C30/0xF0C31             | Data write / last-written read and ready-status read                                               |
| \$F0/\$F1                     | PSG       | 0xF0C00                     | Register select / data                                                                             |
| \$F2/\$F3                     | TED       | 0xF0F00 / 0xF0F20 - 0xF0F6B | Register select / data (audio indices \$00-\$05, video indices \$20-\$32 x4 stride)                |
| \$FE/\$FD/\$BE/\$FA/\$FB/\$FC | ULA       | 0xF2000-0xF2014             | Border, control, status, VRAM address latch low/high, and paged VRAM data                          |

## Z80 16-bit Memory Translation

Z80 memory accesses in 0xF000-0xFFFF are translated to bus 0xF0000-0xF0FFF. ANTIC/GTIA access is provided through the Z80 port bridge (\$D4-\$D7) rather than general memory-mapped addressing.

## x86 Port I/O Bridge

x86 shares the Z80 Voodoo port mapping (\$B0-\$B7) and most sound-chip port mappings, but **POKEY uses direct port-to-register mapping** (not the Z80's select/data protocol). x86 does not implement standard PC VGA I/O ports; VGA access is through the shared bus MMIO aperture and the direct \$A0000-\$AFFFF VRAM memory window.

| x86 Port  | Chip      | Bus Target | Protocol                  |
|-----------|-----------|------------|---------------------------|
| \$B0-\$B7 | Voodoo 3D | 0xF8000    | Addr lo/hi + 4 data bytes |

| x86 Port  | Chip  | Bus Target                  | Protocol                                                                            |
|-----------|-------|-----------------------------|-------------------------------------------------------------------------------------|
| \$60-\$69 | POKEY | 0xF0D00+(port-0x60)         | <b>Direct</b> - port offset maps to writable register                               |
| \$D4/\$D5 | ANTIC | 0xF2100                     | Register select / data (x4 stride)                                                  |
| \$D6/\$D7 | GTIA  | 0xF2140                     | Register select / data (x4 stride, collision regs through 0xF21F8)                  |
| \$E0/\$E1 | SID   | 0xF0E00/0xF0E30/0xF0E50     | Register select / data; select bits 5-6 choose SID1/SID2/SID3                       |
| \$F0/\$F1 | PSG   | 0xF0C00                     | Register select / data                                                              |
| \$F2/\$F3 | TED   | 0xF0F00 / 0xF0F20 - 0xF0F6B | Register select / data (audio indices \$00-\$05, video indices \$20-\$32 x4 stride) |
| \$FE      | ULA   | 0xF2000                     | Border colour only (bits 0-2)                                                       |

**Key difference from Z80:** x86 POKEY access is direct - ports \$60-\$69 map one-to-one onto writable POKEY registers at 0xF0D00+(port-0x60). ANTIC (\$D4/\$D5) and GTIA (\$D6/\$D7) keep their own select/data pairs.

x86 also directly accesses VGA VRAM at \$A0000-\$AFFFF in the memory path (no port translation needed).

## Bank Windows (Z80 / 6502 / x86)

The banked 6502/Z80 ABI uses bank windows to access a 32 MiB banked-CPU visible ceiling from a 16-bit address space; bank translation rejects addresses at or above that ceiling. x86 ordinary addressing remains flat 32-bit, but the x86 bus adapter also implements the same compatibility bank windows plus a VRAM bank window. Those x86 window translations are capped at the legacy DEFAULT\_MEMORY\_SIZE ceiling; ordinary x86 addressing remains clamped by the active visible RAM exposed through the shared bus.

| CPU Address   | Size  | Purpose                            | Bank Select Register             |
|---------------|-------|------------------------------------|----------------------------------|
| \$2000-\$3FFF | 8KB   | Bank 1 (sprite data)               | \$F700/\$F701 (lo/hi)            |
| \$4000-\$5FFF | 8KB   | Bank 2 (font data)                 | \$F702/\$F703 (lo/hi)            |
| \$6000-\$7FFF | 8KB   | Bank 3 (general)                   | \$F704/\$F705 (lo/hi)            |
| \$8000-\$BFFF | 16KB  | VRAM window                        | \$F7F0 (bank number)             |
| \$F000-\$FFF9 | 4090B | I/O window, excluding 6502 vectors | Hardwired: \$Fxxx -> bus \$F0xxx |
| \$F200-\$F24F | 80B   | Coprocessor gateway subrange       | Hardwired: -> bus \$F2340+offset |
| \$FFFA-\$FFFF | 6B    | 6502 vectors (NMI/RESET/IRQ)       | Identity mapped                  |

## 6502 I/O Chip Page Dispatch

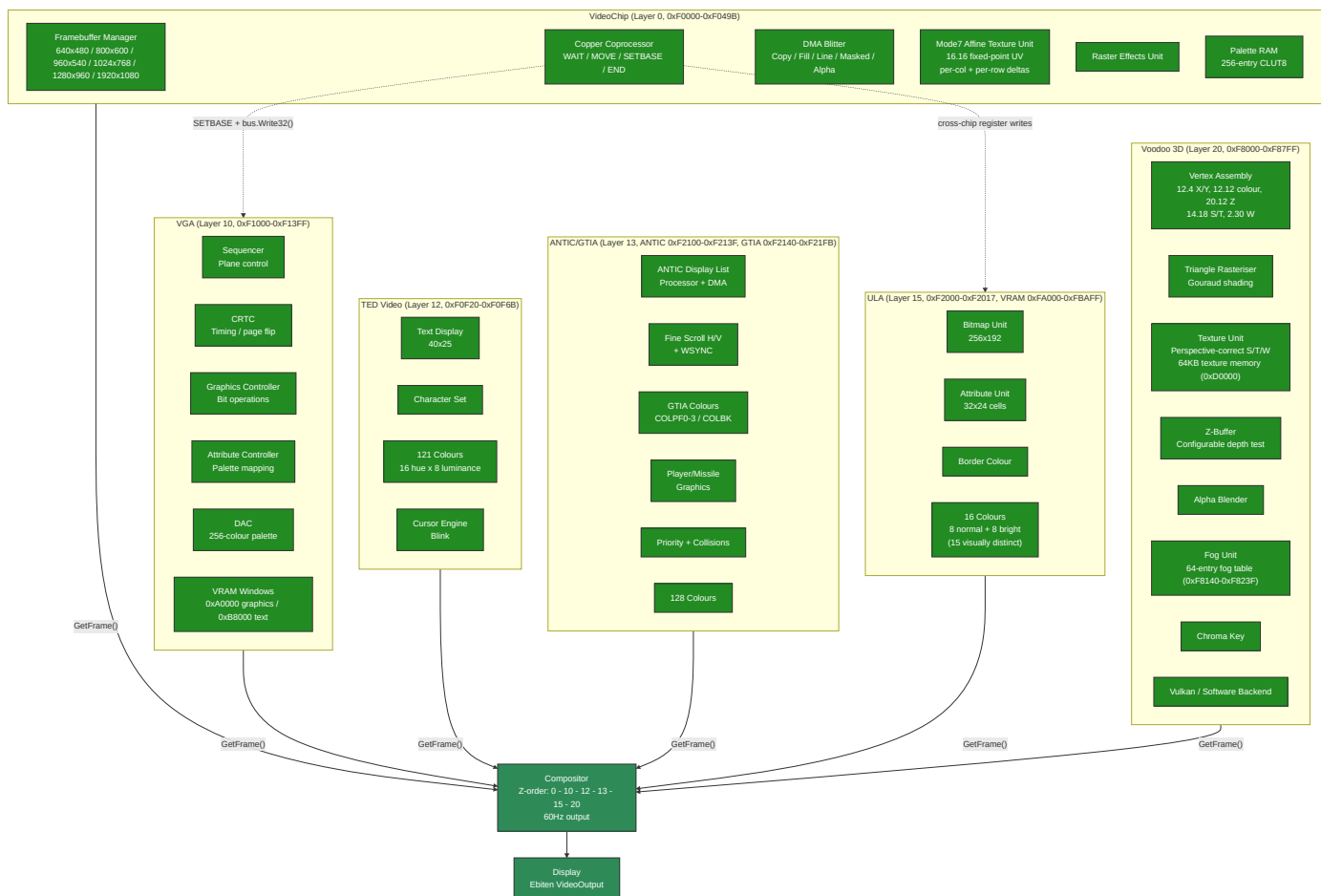
The 6502 uses ioTable[page] to route memory-mapped I/O through the bus:

| 6502 Address  | Chip  | Bus Target     | Notes          |
|---------------|-------|----------------|----------------|
| \$D200-\$D209 | POKEY | 0xF0D00+offset |                |
| \$D400-\$D40F | PSG   | 0xF0C00+offset |                |
| \$D500-\$D51F | SID1  | 0xF0E00+offset | 32-byte window |
| \$D520-\$D53F | SID2  | 0xF0E30+offset | 32-byte window |
| \$D540-\$D55F | SID3  | 0xF0E50+offset | 32-byte window |

| 6502 Address  | Chip      | Bus Target        | Notes                                                            |
|---------------|-----------|-------------------|------------------------------------------------------------------|
| \$D600-\$D605 | TED Audio | 0xF0F00+offset    |                                                                  |
| \$D620-\$D632 | TED Video | 0xF0F20+offset x4 | Stride-4 register mapping including raster compare registers     |
| \$D700-\$D70D | VGA       | 0xF1000           | Direct handler call plus DAC read index, DAC mask, and VRAM bank |
| \$D800-\$D817 | ULA       | 0xF2000+offset    | Registers plus paged VRAM data port                              |

ANTIC/GTIA intentionally has no 6502 \$D400/\$D000 compatibility surface; \$D400-\$D40F is PSG on the 6502 map.

## 4. Video Subsystem



VideoChip routes 32-bit VRAM writes before the register switch, so ordinary framebuffer stores avoid the control-register decode path. Rectangle-shaped blitter fills in raster-op modes update the front buffer directly and mark the covered dirty tiles once when the destination stride is the display row stride; non-rectangular strides keep the per-pixel path for exact dirty tracking. Raster band writes through bus memory aggregate backing invalidation per row.

## Voodoo 3D State Binding

Voodoo triangle submission is state-machine driven. A write that commits `V00D00_TRIANGLE_CMD` binds the current Voodoo raster state to that triangle: `V00D00_FBZ_MODE`, `V00D00_ALPHA_MODE`, `V00D00_FBZCOLOR_PATH`, `V00D00_TEXTURE_MODE`, fog state, chroma key, stipple, clip rectangle, slope registers, and the currently uploaded texture. Later register writes or texture uploads do not affect already submitted triangles. `V00D00_SWAP_BUFFER_CMD` may flush the queued batch later, but the batch is rasterised in triangle submission order using each triangle's bound state.



Voodoo texture slots retain immutable uploaded textures by slot identifier; `V00D00_TEX_BIND` selects a resident texture for subsequently submitted triangles without another guest-memory transfer, and `SYSINFO` advertises the slot contract. Write the slot identifier to `V00D00_TEX_SLOT` before `V00D00_TEX_UPLOAD` to retain that upload. Binding an empty or out-of-range slot leaves the current texture unchanged. The `SYSINFO_FEATURE_V00D00_TEX_SLOTS` bit identifies machines that implement these registers.

When `IE_SWAP_HASH=1`, each presented `V00D00_SWAP_BUFFER_CMD` receives a deterministic sequence number and a 32-bit FNV-1a frame hash captured immediately after readback. `V00D00_SWAP_HASH_SEQ` (0xF8358) returns the latest sequence; write a sequence to `V00D00_SWAP_HASH_QUERY` (0xF835C) and read `V00D00_SWAP_HASH_VALUE` (0xF8360) to retrieve it. The 64-entry history returns zero for unknown or evicted sequences. Hashing is disabled by default, but swap sequence numbering still advances when readback produces no frame.

Consecutive triangles may share an internal state snapshot until a raster-state register or texture upload changes; that sharing is not guest-visible. Fog-table and palette raster lookups remain compatibility-pending. The software backend is the conformance reference; it rasterises each flush from per-flush copies of the bound snapshots under a framebuffer lock, so live raster-state register writes are never blocked by an in-flight raster. Vulkan renders multi-state frames natively by binding each snapshot's pipeline, scissor, push constants, and texture per state group inside one command buffer, and is the only render path in windowed builds: the software rasteriser is the headless build's renderer and the conformance reference, never a runtime fallback. Flushes that arrive without a guest clear render standalone under the clear pass by default; an experimental load-op render pass variant (`IE_VOODOO_LOAD_PASS=1`) composites them over the previous GPU frame. Pipeline or descriptor creation failures drop the frame and log the error rather than silently downgrading.

Voodoo swap jobs run asynchronously; oversized triangle batches render mid-frame without presentation or swap callbacks, and `STATUS` exposes busy and `SWAPBUF` while a presented swap is pending. `V00D00_SWAP_BUFFER_CMD` hands the current batch to the swap worker, and `V00D00_STATUS` reports framebuffer and SST busy plus `SWAPBUF` while that presented swap is active. A later swap waits only when the swap pipeline is full (two jobs in flight), giving up to two frames of run-ahead. If a frame exceeds `V00D00_MAX_BATCH_TRIANGLES`, the full batch is rendered into the draw buffer as a mid-frame render-only flush, without presenting the frame or firing swap callbacks, and triangle submission continues into a fresh batch. This preserves oversized tiled frames instead of dropping triangles.

## Copper Cross-Chip Bus Access

The copper coprocessor is internal to VideoChip but can write to any MMIO-mapped chip on the bus:

- **Default:** `MOVE` targets VideoChip's own registers (`copperIOBase = VIDEO_REG_BASE`)
- **SETBASE opcode** changes `copperIOBase` to any MMIO address (e.g. `VGA_BASE 0xF1000`)
- Subsequent `MOVE` instructions compute `regAddr = copperIOBase + (regIndex * 4)` and route via `bus.Write32()` to VGA, ULA, or any other chip's MMIO
- This enables per-scanline palette changes, mode switches, and register manipulation on any video chip
- `copperIOBase` resets to `VIDEO_REG_BASE` at the start of each frame
- The compositor's `ScanlineAware` interface orchestrates this: `StartFrame()` -> `ProcessScanline(y)` -> `FinishFrame()`

## Extended Blitter: BPP Modes, Draw Modes, Colour Expansion, and Scale

The blitter supports two pixel formats via `BLT_FLAGS` (0xF0488): `RGBA32` (4 bpp, default) and `CLUT8` (1 bpp). Bits 4-7 select one of 16 raster draw modes (Clear, And, Copy, Xor, Invert, etc.) applied per pixel during `FILL` and `COPY` operations. `BLT_OP=4` performs source-over alpha blending with source alpha in bits 31-24 using  $out = (src*a + dst*(255 - a)) / 255$ . With `BLT_FLAGS` bit 12 (`ALPHA_TMPL`) set, `BLT_OP=4` instead treats `BLT_SRC` as an 8 bpp alpha plane (row stride in `BLT_SRC_STRIDE`) and blends the `BLT_FG` colour over the destination with the same source-over equation. This serves antialiased glyph and icon rendering (the AROS driver uses it for `PutAlphaTemplate`). `BLT_OP=7` performs nearest-

neighbour scaling in RGBA32 or CLUT8. When BLT\_FLAGS=0, the blitter defaults to Copy mode with RGBA32 for full backward compatibility.

Masked copies (BLT\_OP=3) copy source pixels only where a 1-bit mask is set. The mask base address is in BLT\_MASK, the mask row stride in BLT\_MASK\_MOD (0 selects packed rows of (width+7)/8 bytes), and BLT\_MASK\_SRCX gives the starting bit offset within each mask row. Mask bits are sampled LSB-first by default; setting BLT\_FLAGS bit 11 (MASK\_MSB) selects MSB-first sampling, which matches the Amiga PLANEPTR convention used by AROS layer masks, so the driver can pass CopyBoxMasked masks to the hardware without conversion.

BLT\_CTRL bit 0 starts the synchronous blit, bit 1 is read-only busy, and bit 2 enables a completion pulse on IntMaskBlitter. The BLT\_CTRL start edge samples the register shadow from the shared bus image in little-endian register order; VideoChip big-endian mode affects guest-facing register and pixel access, not the internal bus-shadow hydration order. BLT\_STATUS bit 0 is ERR, bit 1 is DONE, and bit 2 is sticky IRQ\_PENDING (write 1 to clear). Invalid opcodes, out-of-range Mode7 samples, overflowed blitter bounds, and destination rectangles outside CPU-visible writable memory set ERR and do not silently fall back to COPY or wrap into another address range.

The colour expansion operation (BLT\_OP=6) renders 1-bit glyph templates into coloured pixels for hardware-accelerated text. It reads a template from BLT\_MASK, uses BLT\_FG/BLT\_BG (0xF048C/0xF0490) as foreground/background colours, and supports three modes: JAM2 (opaque - set bits write FG, clear bits write BG), JAM1 (transparent - only set bits write FG), and Invert (set bits XOR the destination). BLT\_MASK\_MOD (0xF0494) sets the template row stride and BLT\_MASK\_SRCX (0xF0498) provides sub-byte bit alignment for glyph fragments. Template bits are MSB-first (Amiga convention).

Line drawing (BLT\_OP=2) supports an extended mode when BLT\_FLAGS != 0: BLT\_DST becomes the framebuffer base address, BLT\_WIDTH holds the packed endpoint coordinates (y1<=16) | x1, and BLT\_DST\_STRIDE sets the row stride. This allows line drawing into arbitrary bitmaps (not just the active framebuffer) with BPP awareness and all 16 draw modes. When BLT\_FLAGS=0, legacy behaviour is preserved (endpoint in BLT\_DST, base at VRAM\_START). In extended mode the blitter does not clip - callers must provide pre-clipped coordinates (the AROS driver uses Cohen-Sutherland clipping before calling the blitter).

Scale blits (BLT\_OP=7) use BLT\_SRC/BLT\_DST as base addresses, BLT\_WIDTH/BLT\_HEIGHT as the source size in pixels, and BLT\_COLOR as a packed destination size (height <= 16 | width). BLT\_SRC\_STRIDE defaults to source width times bytes-per-pixel. BLT\_DST\_STRIDE defaults to the active VideoChip mode stride for VRAM destinations or destination width times bytes-per-pixel for ordinary memory. In non-direct VRAM mode, visible VRAM accesses use the active VideoChip front buffer and off-screen VRAM aperture accesses fall back to bus memory so back buffers can be presented through VIDEO\_FB\_BASE. Out-of-bounds source or destination rectangles, rectangles crossing the VRAM aperture end, zero source dimensions, and zero destination dimensions set BLT\_STATUS.ERR.

Raster-band register draws (VIDEO\_RASTER\_Y, VIDEO\_RASTER\_HEIGHT, VIDEO\_RASTER\_COLOR, VIDEO\_RASTER\_CTRL) target the active framebuffer source. In RGBA32 mode, a nonzero VIDEO\_FB\_BASE writes the band into bus memory at that base; otherwise it writes the internal front buffer or direct VRAM slice. In CLUT8 mode, raster bands write indexed bytes at VIDEO\_FB\_BASE.

| Register      | Address | Description                                                                                                                          |
|---------------|---------|--------------------------------------------------------------------------------------------------------------------------------------|
| BLT_FLAGS     | 0xF0488 | BPP (bits 0-1), draw mode (bits 4-7), JAM1/invert flags (bits 8-10), MSB-first mask sampling (bit 11), alpha-template blend (bit 12) |
| BLT_FG        | 0xF048C | Foreground colour for colour expansion                                                                                               |
| BLT_BG        | 0xF0490 | Background colour for colour expansion                                                                                               |
| BLT_MASK_MOD  | 0xF0494 | Template row modulo (bytes per row)                                                                                                  |
| BLT_MASK_SRCX | 0xF0498 | Starting X bit offset in template                                                                                                    |

## Mode7 Affine Texture Unit

The blitter's Mode7 operation (`blt0pMode7`) implements SNES-style affine texture mapping as a DMA blitter mode within VideoChip. It uses 8 dedicated MMIO registers (`0xF0058-0xF0074`):

| Register         | Address | Description                                      |
|------------------|---------|--------------------------------------------------|
| BLT_MODE7_U0     | 0xF0058 | Starting U coordinate (16.16 fixed-point)        |
| BLT_MODE7_V0     | 0xF005C | Starting V coordinate (16.16 fixed-point)        |
| BLT_MODE7_DU_COL | 0xF0060 | U increment per column (16.16)                   |
| BLT_MODE7_DV_COL | 0xF0064 | V increment per column (16.16)                   |
| BLT_MODE7_DU_ROW | 0xF0068 | U increment per row (16.16)                      |
| BLT_MODE7_DV_ROW | 0xF006C | V increment per row (16.16)                      |
| BLT_MODE7_TEX_W  | 0xF0070 | Texture width mask (must be power-of-2 minus 1)  |
| BLT_MODE7_TEX_H  | 0xF0074 | Texture height mask (must be power-of-2 minus 1) |

The rasteriser walks each destination pixel, computes the source UV from the affine matrix (origin + column delta + row delta), wraps via power-of-2 bitmask, and samples the source texture. This enables rotation, scaling, and perspective-like effects on tiled backgrounds - the same technique used by the SNES PPU2 for its Mode 7 background layer.

VideoChip Mode7 honours the `BLT_FLAGS BPP` field: RGBA32 samples and writes 4-byte pixels, while CLUT8 samples and writes 1-byte palette indices with BPP-aware default strides. In RGBA32 mode the default source stride is  $(\text{texture mask width} + 1) * 4$ ; in CLUT8 mode the default source stride is  $(\text{texture mask width} + 1)$ . The destination stride follows the same BPP-aware default-stride rules as the other blitter operations: active framebuffer width in bytes for VRAM destinations, or destination width times bytes-per-pixel for ordinary memory.

## Video Compositor

The compositor collects immutable frame snapshots from all enabled video sources and blends them in Z-order (layer 0 at the back, layer 20 at the front). For IEVideoChip CLUT8 mode, mapped VRAM and direct bus-backed VRAM are converted through the palette before compositing. CLUT8 conversion is cached only when `fbBase` points at the VideoChip-visible VRAM window; direct RAM-backed framebuffer modes are converted every frame because guest CPU stores can bypass VideoChip dirty tracking. Desktop startup uses a 1920x1080 fullscreen presentation by default, and the x64 live-image launcher locks it fullscreen through `IE_LIVE_IMAGE=1`. Native sources keep their requested dimensions. The default native mode is 960x540 for exact 2x 1080p presentation. Video compositor default scale mode is stretch-fill; F11 toggles non-16:9 sources to aspect-fit. Shift+F11 toggles fullscreen/windowed mode when that mode is not locked.

Two rendering paths:

1. **Scanline-aware path** - used when at least one enabled source implements `ScanLineAware`. The compositor advances scanline-capable sources in sorted layer order for each scanline, then blends all enabled sources in the global layer order. If exactly one source also implements `ScanLineBatchAware`, it advances the whole scanline range under one source lock. Multi-source scanline composition keeps per-scanline interleaving because that ordering is guest-visible.
2. **Full-frame fallback** - used when no enabled source is scanline-aware. It collects complete frames and blends them in sorted layer order. Same-size frame blending uses persistent strip workers instead of spawning goroutines each frame. Sources can optionally report dirty rectangles; outputs that implement `RegionUpdatingOutput` receive region uploads, while other outputs keep the full-frame upload path. Sources marked `OpaqueFrameSource` can use a promotion-aware copy path.

All-zero frame pixels are transparent; any nonzero alpha or RGB value is opaque. During compositing, zero-alpha nonzero-RGB pixels are promoted to opaque 0xFFRRGGBB before they overwrite the destination. The compositor tick remains fixed at 60 Hz for guest VBlank compatibility; `GetRefreshRate()` reports the output backend rate, while `GetTickRate()` reports the compositor tick.

`FrameGenerationSource` lets the compositor skip collect/copy/blend/upload work only after source `TickFrame` hooks run and only when every enabled source generation is unchanged. A pending full-frame update, a frame-complete callback, an enabled source without generation tracking, or a scanline-compositing source disables this skip for that tick.

Video frame leases are enabled by default and can be disabled with `IE_VIDEO_FRAME_LEASES=0`. Copy-buffer paths use three-slot lease rings to keep source-layer buffers alive while the hardware compositor or snapshot path still references them. Sources that implement the compositor copy interface can copy a stable frame directly into the caller-provided lease buffer, avoiding an intermediate source snapshot before compositor handoff. The software output path can also retain a lease-backed final frame for `UpdateFrame` instead of copying through the legacy output buffer. Frame leases keep compositor handoff buffers stable until release; hardware layers retain leases or stage copies when leases are unavailable. Snapshot APIs still return deep copies.

SIMD acceleration kernels are enabled by default on amd64 builds and can be disabled with `IE_SIMD=0`. Hot pixel and sample spans (compositor blend and normalise, blitter fill and colour expand, software Voodoo untextured spans, audio post-effects) dispatch through package-level function variables that point at bit-exact SIMD variants when the host provides the AVX2 baseline; every kernel keeps a scalar leaf as its canonical reference, and non-amd64 targets, hosts without the baseline, or `IE_SIMD=0` route the scalar path. The SIMD variants are additive and never alter emulated results.

## Triple-Buffer Protocol

All video systems except `VideoChip` use a lock-free triple-buffer protocol for `GetFrame()`:

```
Slots:  writeIdx = 0 (producer-owned)
        sharedIdx = 1 (atomic, in-transit)
        readingIdx = 2 (consumer-owned)

Producer (render goroutine, after rendering into frameBufs[writeIdx]):
    writeIdx = sharedIdx.Swap(writeIdx)

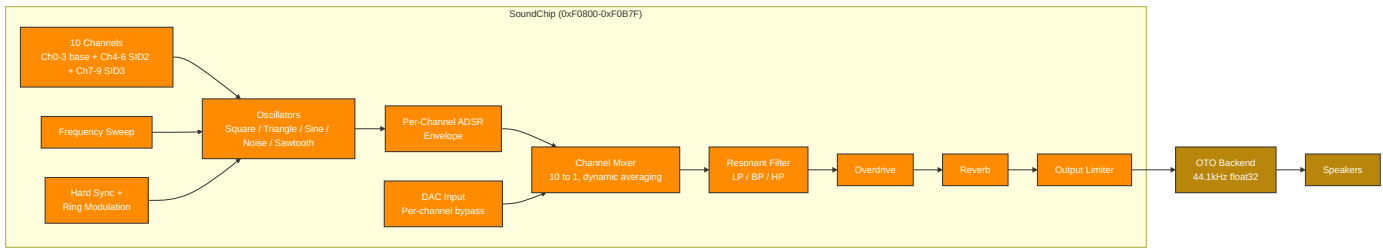
Consumer (compositor, GetFrame):
    readIdx = sharedIdx.Swap(readIdx)
    return frameBufs[readIdx]
```

**Important:** `GetFrame()` performs an atomic `Swap` - calling it twice in a row swaps back to the previous buffer. Always call once and save the result.

On resolution change, all 3 buffer slots are reallocated and indices reset to `writeIdx=0`, `sharedIdx=1`, `readingIdx=2`.

# 5. Audio Subsystem

## Synthesis Pipeline



## SoundChip Dual-Interface Architecture

The SoundChip exposes two register interfaces for its channels:

**Legacy per-waveform interface** (0xF0900-0xF09FF) - dedicated register blocks with two decoded sweep-register exceptions:

Decoded ranges: 0xF0900-0xF093F except 0xF0914 and 0xF0918, 0xF0940-0xF097F plus 0xF0914, and 0xF0980-0xF09BF plus 0xF0918.

| Range                        | Channel | Default Waveform                                                   |
|------------------------------|---------|--------------------------------------------------------------------|
| 0xF0900-0xF093F              | Ch 0    | Square, except 0xF0914 is triangle sweep and 0xF0918 is sine sweep |
| 0xF0940-0xF097F plus 0xF0914 | Ch 1    | Triangle                                                           |
| 0xF0980-0xF09BF plus 0xF0918 | Ch 2    | Sine                                                               |
| 0xF09C0-0xF09FF              | Ch 3    | Noise                                                              |
| 0xF0A00-0xF0A6F              | -       | Sawtooth + modulation/effects                                      |

Within the modulation/effects window, 0xF0A00-0xF0A1C holds the sync-source and ring-modulation-source registers, 0xF0A20-0xF0A6F holds the sawtooth legacy registers, and 0xF0A40, 0xF0A50, and 0xF0A54 hold overdrive and reverb controls.

**FLEX unified interface** (0xF0A80-0xF0B7F) - 4 channels with identical 64-byte register blocks. Each channel can be any waveform type:

| Offset     | Register        | Description                                                    |
|------------|-----------------|----------------------------------------------------------------|
| +0x00      | FREQ            | 16.8 fixed-point Hz (value / 256.0)                            |
| +0x04      | VOL             | Volume (0-255)                                                 |
| +0x08      | CTRL            | Enable (bit 0), gate (bit 1)                                   |
| +0x0C      | DUTY            | Pulse width duty cycle; high byte is PWM depth scaled by 1/256 |
| +0x10      | SWEEP           | Frequency sweep rate                                           |
| +0x14-0x20 | ATK/DEC/SUS/REL | ADSR envelope                                                  |
| +0x24      | WAVE_TYPE       | Waveform selection                                             |
| +0x28      | PWM_CTRL        | Pulse width modulation                                         |
| +0x2C      | NOISEMODE       | Noise mode (0=white, 1=periodic, 2=metallic, 3=psg)            |

| Offset | Register | Description                                                 |
|--------|----------|-------------------------------------------------------------|
| +0x30  | PHASE    | Phase reset                                                 |
| +0x34  | RINGMOD  | Ring modulation (bit 7=enable, 0-2=source)                  |
| +0x38  | SYNC     | Hard sync (bit 7=enable, 0-2=source)                        |
| +0x3C  | DAC      | DAC mode bypass (signed 8-bit sample, -128=-1.0, +127=+1.0) |

FLEX channels are at FLEX\_CH0\_BASE = 0xF0A80, stride = 0x40. Primary channels 0-3 occupy 0xF0A80-0xF0B7F; SID2 flex channels 4-6 occupy 0xF0C40-0xF0CFF; SID3 flex channels 7-9 occupy 0xF0D40-0xF0DFF. AUDIO\_CTRL uses bit 0 for enable and bit 1 for freeze. ENV\_SHAPE remains at 0xF0804 for channel 0, with per-channel shapes at 0xF0860 + channel\*4. Both legacy and FLEX interfaces write to the same underlying 10-channel mixer. The FLEX interface is preferred for new code - the legacy interface exists for backward compatibility.

## Filter and Modulation

The SoundChip's global resonant filter (0xF0820-0xF0830) supports low-pass, band-pass, and high-pass modes with cutoff frequency, resonance, and optional filter modulation source/amount registers. Cutoff and resonance smooth toward register targets at the same 0.02 coefficient used by per-channel filters. SID 12-bit DAC quantization truncates to remove half-LSB DC bias.

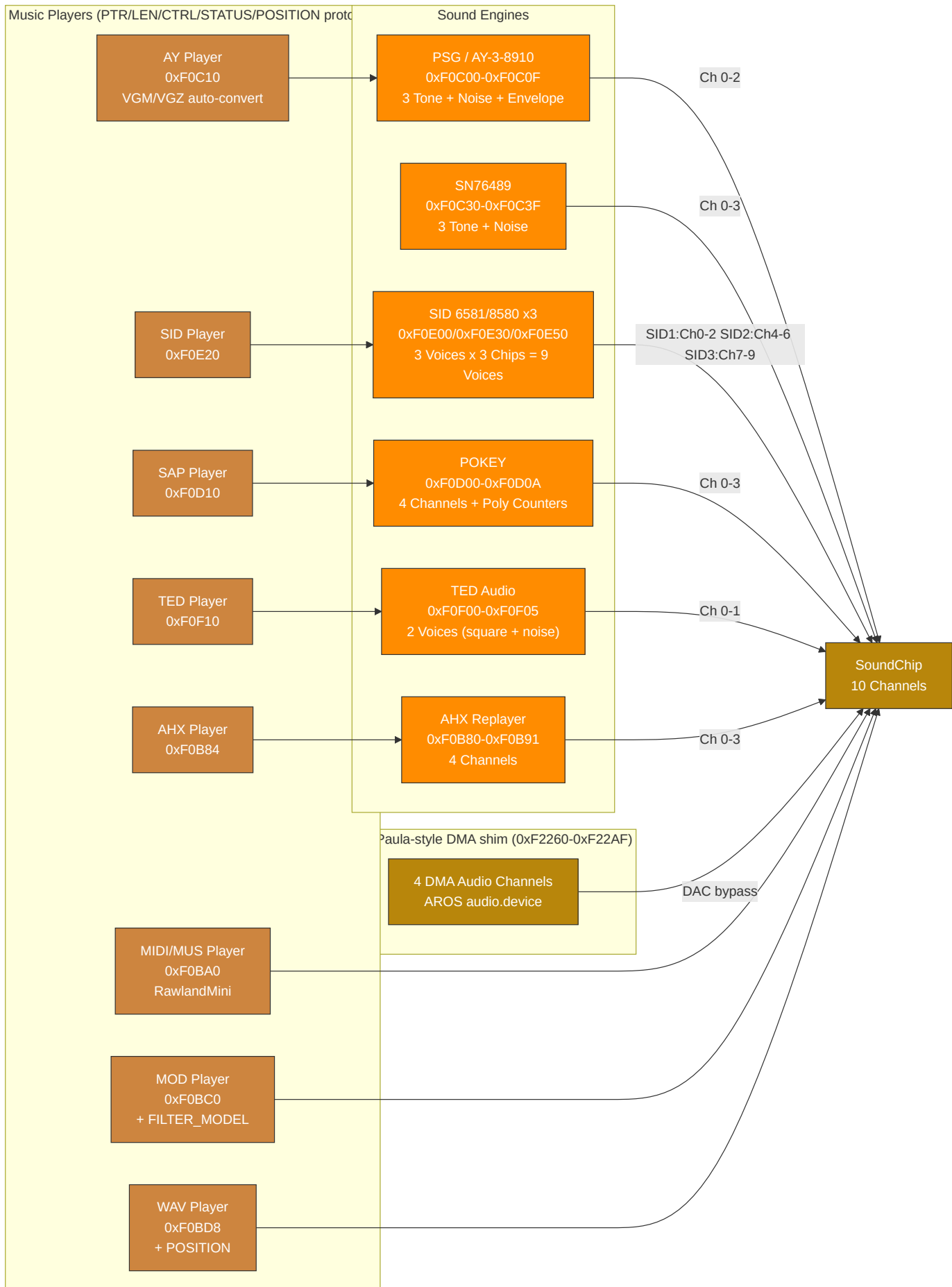
## Sample Generation Locking

ReadSample advances registered sample tickers before mixing the same sample. GenerateSample then takes chip.mu only for channel-state mutation, channel mixing, register-visible smoothing state, and a post-processing configuration snapshot. SID mixer options, overdrive, the global filter, reverb, and the master normaliser run after chip.mu is released. Their mutable DSP state is protected by postFXMu, which is also used by reset, debug snapshot, and restore.

ReadSamples(dst) is the block-rendering API used by the OTO backend. It installs a pending audio-block state for the duration of the pull. ReadSamples uses safe block ticking only when every active sample ticker implements ReadSamplesBlockTicker, SFX allows block ticking, and no sample mixers are registered. In that case it advances eligible tickers in bounded blocks and flushes each block immediately. Otherwise it records each sample's ticker output and mixer contribution in order, then flushes pending segments of up to 64 samples under one channel-mix lock. Any register write or SoundChip setter reached from a ticker first flushes the pending segment, so single-threaded ReadSamples output matches the ReadSample loop. Concurrent CPU writes remain bounded by the segment size and are allowed to apply earlier than a sample-at-a-time interleaving.

The master normaliser separates setter-owned configuration from audio-thread-owned dynamics. Configuration setters publish a generation counter; the audio thread applies pending envelope and lookahead resets before the next normaliser step. Register-driven engine synchronisation can batch SoundChip writes through HandleRegisterWrites, preserving write order while taking the chip mutex once. Idle playback paths for SID, TED, MIDI, MOD, AHX, POKEY, and SFX use atomic gates before taking their engine or channel mutexes. POKEY event playback drains current-sample events through a reusable scratch buffer prepared when events are loaded.

## Engine and Player Routing





## Audio Engine Plus Enhanced Mode

All five classic-style sound engines have a "Plus" enhanced mode, activated by writing 1 to their respective PLUS\_CTRL register:

| Engine | PLUS_CTRL Address | Enhancements                                                                      |
|--------|-------------------|-----------------------------------------------------------------------------------|
| PSG+   | 0xF0C20           | Enhanced render path: oversampling, filtering, drive/room shaping, stereo voicing |
| SID+   | 0xF0E19           | Enhanced render path for SID voices (oversampling/filter/drive/room shaping)      |
| POKEY+ | 0xF0D09           | Enhanced render path (oversampling/filter/drive/room shaping)                     |
| TED+   | 0xF0F05           | Enhanced render path plus TED-specific response shaping                           |
| AHX+   | 0xF0B80           | AHX voice-state mapping with stereo spread/panning and room processing            |

The PSG uses the AY/YM logarithmic 16-step volume curve by default. A legacy linear curve is retained for older material that depends on it.

When Plus mode is enabled, the engine retains full backward compatibility with the standard register set while exposing additional capabilities. AHX maps tracker state to native SoundChip channels instead of producing an AROS audio DMA stream; AHX+ uses a 64-sample crossfade when enabling/disabling to prevent audio glitches.

## AROS Paula-Style DMA Shim ABI

The AROS audio block at 0xF2260-0xF22AF is an Intuition Engine shim for AROS `audio.device`, not a full Amiga Paula implementation.

- Guest code writes the shim with 32-bit aligned writes (`MOVE.L`). Narrower writes are outside the current ABI.
- A DMACON channel transition from 0 to 1 latches that channel's pointer, length, period, and volume. Active playback uses the latched values until the buffer ends.
- If the latched period or length is zero at arm time, the channel is not activated. The channel DMACON bit is cleared, the status bit is set, and a level-3 interrupt is raised when the corresponding INTENA bit is set.
- Buffer exhaust sets the status bit, raises a level-3 interrupt when the corresponding INTENA bit is set, marks the channel inactive, and clears that channel's DMACON bit. The guest must write DMACON enable again to start the next buffer.
- Out-of-range pointers beyond the active AROS profile RAM deactivate the channel, mute the DAC output, set the status bit, and raise a level-3 interrupt when the corresponding INTENA bit is set.
- Pointer writes ignore bit 0, length is a word count and preserves odd values, period writes with zero are ignored, and volume writes are clamped to 0..64.

## AROS HostFS DOS Handler ABI

The AROS DOS block at 0xF2220-0xF225F is the MMIO command bridge used by the AROS m68k-ie packet handler. The guest writes up to four argument registers and then writes `AROS_DOS_CMD`; the Go-side `ArosDOSDevice` executes the request synchronously and returns AmigaDOS-style `RESULT1` and `RESULT2` values. `ADOS_CMD_EXAMINE_ALL` accelerates `ACTION_EXAMINE_ALL` through a 20-byte big-endian request descriptor, guest span validation, direct `ExAllData` packing, `eac_LastKey` continuation, and `ERROR_ACTION_NOT_KNOWN` fallback for match strings or hooks.

AROS HostFS fast paths use strict bulk guest-memory helpers, a 64 KiB sequential read-ahead cache, and cache invalidation on non-sequential reads, seeks outside cache, writes, truncates, close, create, delete, rename, and dirty close paths. Spans outside active guest RAM, high-pointer requests unsupported by the active bus, and low/high backing seam crossings fail closed. External host changes are best-effort and are not continuously watched.

## AROS Host Socket ABI

The AROS host socket block at 0xF2500-0xF257F backs the m68k-ie ROM `bsdsocket.library`. 0xF2400-0xF24FF remains the SYSINFO ABI, so socket registers occupy the next 128-byte MMIO block. The AROS host socket block at 0xF2500-0xF257F uses 96-byte big-endian request descriptors, strict guest-memory helper copies, 64 KiB send/rcv caps, 128-byte sockaddr caps, guest-visible descriptor handles, `WaitSelect` `fd_set` translation, `Release/ReleaseCopy/Obtain` transfer keys, and `ENOSYS` fail-closed behaviour when disabled. The guest writes `REQ_PTR` and `REQ_LEN`, places the descriptor in guest RAM, and triggers synchronous dispatch by writing `CMD`. Payload, `fd_set`, `timeval`, `hostname`, `hostent`, and `sockaddr` buffers are copied through those validated guest spans.

Supported command values are 1=`socket`, 2=`bind`, 3=`listen`, 4=`accept`, 5=`connect`, 6=`sendto`, 7=`recvfrom`, 8=`shutdown`, 9=`setsockopt`, 10=`getsockopt`, 11=`getsockname`, 12=`getpeername`, 13=`ioctl`, 14=`close`, 15=`WaitSelect`, 16=`gethostbyname`, 17=`gethostbyaddr`, 18=`gethostname`, 19=`Dup2`, 20=`GetEvents`, 21=`ReleaseSocket`, 22=`ReleaseCopyOfSocket`, and 23=`ObtainSocket`. The Unix backend supports IPv4 TCP and UDP sockets, makes host file descriptors non-blocking, hides them behind guest-visible handles, maps `WaitSelect` `fd_sets` through that handle table, and rejects arbitrary emulator process descriptors with `EBADF`. `ReleaseSocket` removes the caller's guest descriptor and stores the host descriptor under a release key. `ReleaseCopyOfSocket` stores a duplicated host descriptor and leaves the caller's descriptor active. Passing release key -1 asks the backend to allocate a positive key; `ObtainSocket` consumes that key and returns a new guest descriptor, or fails with `EWouldBlock` when the key is absent. Raw sockets, packet filters, route manipulation, interface configuration, and monitoring APIs return unsupported errors. When networking is disabled or no backend is present, the block fails closed with `ENOSYS`.

## Subsong Selection

SID, SAP, and AHX players support subsong selection for multi-tune files. Each player has a subsong register that selects which tune to play from a multi-song file.

SID PSID playback captures CIA1 timer-A latch writes at \$DC04/\$DC05; when non-zero, the player uses that latch as `cyclesPerTick`, so multispeed tunes run at `clockHz / latch`. SID MMIO opts into wide-write fanout: 16-bit and 32-bit writes are decomposed into little-endian byte register writes.

## Generic Live-MIDI Port (Shared Subsystem)

The live-MIDI port is a CPU-agnostic MMIO device that feeds a raw running-status MIDI byte stream straight into the synth, distinct from the file player (`SOUND PLAY "x.mid"`) but sharing the same `MIDIEngine`, the same pool of 10 MIDI voices, and the same sound-chip mix key. There is one synth and one runtime-status owner; the live port and the file player never double-mix.

Three byte-wide registers drive it: `IE_MIDI_LIVE_DATA` (0xF0BF4, write a raw MIDI byte), `IE_MIDI_LIVE_STATUS` (0xF0BF5, read bit 0 = live port active), and `IE_MIDI_LIVE_CTRL` (0xF0BF6, write bit 0 = reset / all notes off).

`midi_live.go` is a running-status state machine that turns the byte stream into `MIDIEvents` (note-on/off, control change, program change, pitch bend, including messages split across multiple writes) and applies them to the shared engine. Because the device hangs off the `MachineBus`, every core and `EhBASIC` reach it directly; the `EhBASIC` `MIDI NOTE/PROG/CTRL/SEND/RESET` keywords emit onto these registers. Live notes take precedence over the file player at voice allocation while active (a live note steals a file voice before another live voice) and clear on a `CTRL` reset. Per-architecture symbol names are declared in each SDK include (`ie32/ie64/ie65/ie68/ie80/ie86.inc`), windowed to each core's address convention. The Ebiten runtime status bar's MIDI legend lights when either the file player is playing or the live port is active (`MIDIEngine.LiveActive()`), so both MIDI sources share one indicator.

## EmuTOS Atari MIDI ACIA bridge

EmuTOS does not know IE's native live-MIDI register map; it talks MIDI only through the Atari ST MIDI port, an **MC6850 ACIA**. `atari_midi_acia.go` is a minimal, output-only MC6850 shim (AtariMIDIACIA) that bridges that ACIA to the same LiveMIDI/MIDIEngine path, so notes from ST software or GEMDOS `Bconout(3, ...)` reach IE's synth. It is wired **only in EmuTOS mode** (`main.go`, guarded by `modeEmuTOS`) and holds no synth or voice state of its own: control writes forward to the LiveMIDI parser, status reads always report TDRE (ready to transmit), and there is no MIDI-in. The shim maps two address forms of the Atari contract `$FFFC04` (RS=0: control/status) and `$FFFC06` (RS=1: TX data / RX): the bus-canonical low-16 alias `0x0000FC04/06` (which the sign-extended guest access `0xFFFFFC04/06` is normalized to before handler lookup) and the 24-bit Atari hardware alias `0x00FFFC04/06` (which does not pass through sign-extension normalization and must be mapped explicitly). Both aliases use `MapI0NoShadow` because they fall inside guest RAM and status/data accesses must not mirror handler values into guest memory. A control write of `ACIA_CTRL_MASTER_RESET` (CR1:CR0 == 11) calls `LiveMIDI.Reset()`. The IKBD ACIA at `$FFFC00/02` is deliberately **not** mapped: keyboard already flows through the IOREC pump in `emutos_loader.go`. This bridge is EmuTOS-specific; there is no generic ACIA emulation for other modes.

## 6. Memory Map

All CPU cores observe the same guest physical address space. Address ranges in this table are not per-core private maps unless explicitly stated. Some ranges have multiple architectural uses: for example, a decoded RAM range may also contain conventional worker buffers, framebuffer memory, or OS/profile-owned allocations. In those cases, the table describes reservations within the shared physical map, not separate mappings.

The `Decoded as` column describes how the bus routes a physical access. The `Architectural use` and `Owner / lifetime` columns describe the guest contract or current subsystem reservation inside that decoded mapping. Overlapping rows are intentional when a reservation lives inside a broader shared-RAM range.

| Range             | Size  | Decoded as            | Architectural use                                                   | Visible to    | Owner / lifetime                  | Notes                                                                                |
|-------------------|-------|-----------------------|---------------------------------------------------------------------|---------------|-----------------------------------|--------------------------------------------------------------------------------------|
| 0x00000 - 0x9EFFF | 636KB | Shared RAM            | Main low-memory programme/data area                                 | All CPU cores | Guest and boot-profile convention | Includes IE32 vector table base at 0x00000 and programme load convention at 0x01000. |
| 0x9F000 - 0x9FFFF | 4KB   | Shared RAM            | Default stack seed region                                           | All CPU cores | CPU reset convention              | IE32 and IE64 seed stack pointers at 0x9F000; it is still ordinary shared RAM.       |
| 0xA0000 - 0xAFFFF | 64KB  | VGA VRAM window       | VGA graphics aperture                                               | All CPU cores | VGA subsystem                     | x86 can also access this as conventional VGA graphics memory.                        |
| 0xB8000 - 0xBFFFF | 32KB  | VGA text buffer       | VGA text aperture                                                   | All CPU cores | VGA subsystem                     | Conventional text-memory range.                                                      |
| 0xD0000 - 0xDFFFF | 64KB  | Voodoo texture memory | Voodoo texture upload/storage window                                | All CPU cores | Voodoo subsystem                  | Shared physical texture-memory aperture.                                             |
| 0xF0000 - 0xF049B | 1180B | MMIO                  | VideoChip, copper, blitter, palette, and extended blitter registers | All CPU cores | Video subsystem                   | Layer-0 video control and DMA-style graphics operations.                             |

| Range             | Size | Decoded as | Architectural use                                          | Visible to    | Owner / lifetime | Notes                                                                                    |
|-------------------|------|------------|------------------------------------------------------------|---------------|------------------|------------------------------------------------------------------------------------------|
| 0xF0700 - 0xF07FF | 256B | MMIO       | Terminal, serial, mouse, keyboard, and RTC_EP0CH (0xF0750) | All CPU cores | I/O subsystem    | Input, terminal, and clock-facing low MMIO.                                              |
| 0xF0800 - 0xF0B7F | 896B | MMIO       | SoundChip channels, including FLEX                         | All CPU cores | Audio subsystem  | Runtime audio-chip register block.                                                       |
| 0xF0B80 - 0xF0B91 | 18B  | MMIO       | AHX engine/player                                          | All CPU cores | Audio subsystem  | Player-facing register block.                                                            |
| 0xF0BA0 - 0xF0BBF | 32B  | MMIO       | MIDI/MUS player                                            | All CPU cores | Audio subsystem  | Player-facing register block.                                                            |
| 0xF0BC0 - 0xF0BD7 | 24B  | MMIO       | MOD player                                                 | All CPU cores | Audio subsystem  | Player-facing register block.                                                            |
| 0xF0BD8 - 0xF0BF3 | 28B  | MMIO       | WAV player                                                 | All CPU cores | Audio subsystem  | Player-facing register block.                                                            |
| 0xF0BF4 - 0xF0BF6 | 3B   | MMIO       | Generic live-MIDI port                                     | All CPU cores | Audio subsystem  | Byte-wide running-status MIDI stream: data write, active-status read, and reset control. |
| 0xF0C00 - 0xF0C0F | 16B  | MMIO       | PSG engine (AY-3-8910/YM2149 registers)                    | All CPU cores | Audio subsystem  | Register-select/data-compatible PSG block.                                               |
| 0xF0C10 - 0xF0C1F | 16B  | MMIO       | PSG / AY player                                            | All CPU cores | Audio subsystem  | Player-facing register block.                                                            |
| 0xF0C20           | 1B   | MMIO       | PSG+ control                                               | All CPU cores | Audio subsystem  | Extended PSG control byte.                                                               |
| 0xF0C30 - 0xF0C3F | 16B  | MMIO       | Native SN76489 latch/data, ready, and LFSR mode registers  | All CPU cores | Audio subsystem  | Native SN76489-compatible control block.                                                 |
| 0xF0C40 - 0xF0CFF | 192B | MMIO       | SID2 flex-style SoundChip channels 4-6                     | All CPU cores | Audio subsystem  | Multi-SID/FLEX channel range.                                                            |
| 0xF0D00 - 0xF0D0A | 11B  | MMIO       | POKEY engine                                               | All CPU cores | Audio subsystem  | POKEY register block.                                                                    |
| 0xF0D10 - 0xF0D20 | 17B  | MMIO       | SAP player                                                 | All CPU cores | Audio subsystem  | Player-facing register block.                                                            |
| 0xF0D40 - 0xF0DFF | 192B | MMIO       | SID3 flex-style SoundChip channels 7-9                     | All CPU cores | Audio subsystem  | Multi-SID/FLEX channel range.                                                            |
| 0xF0E00 - 0xF0E1C | 29B  | MMIO       | SID1 engine (6581/8580)                                    | All CPU cores | Audio subsystem  | Primary SID register block.                                                              |
| 0xF0E20 - 0xF0E2D | 14B  | MMIO       | SID player                                                 | All CPU cores | Audio subsystem  | Player-facing register block.                                                            |

| Range           | Size  | Decoded as               | Architectural use                       | Visible to    | Owner / lifetime      | Notes                                                                                                                                                         |
|-----------------|-------|--------------------------|-----------------------------------------|---------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xF0E30-0xF0E4C | 29B   | MMIO                     | SID2 engine (6581/8580)                 | All CPU cores | Audio subsystem       | Secondary SID register block.                                                                                                                                 |
| 0xF0E50-0xF0E6C | 29B   | MMIO                     | SID3 engine (6581/8580)                 | All CPU cores | Audio subsystem       | Tertiary SID register block.                                                                                                                                  |
| 0xF0E80-0xF0EFF | 128B  | MMIO                     | SoundChip SFX trigger legacy aliases    | All CPU cores | Audio subsystem       | Channels 0-3 aliases for the trigger-oriented SFX range.                                                                                                      |
| 0xF2600-0xF29FF | 1KB   | MMIO                     | SoundChip SFX trigger extended channels | All CPU cores | Audio subsystem       | Channels 0-31 trigger-oriented SFX range.                                                                                                                     |
| 0xF0F00-0xF0F05 | 6B    | MMIO                     | TED audio engine                        | All CPU cores | Audio subsystem       | TED audio registers.                                                                                                                                          |
| 0xF0F10-0xF0F1F | 16B   | MMIO                     | TED player                              | All CPU cores | Audio subsystem       | Player-facing register block.                                                                                                                                 |
| 0xF0F20-0xF0F6B | 76B   | MMIO                     | TED video                               | All CPU cores | TED video subsystem   | Stride-4 TED video register mapping.                                                                                                                          |
| 0xF3000-0xF6FFF | 16KB  | MMIO / TED VRAM aperture | TED private video RAM                   | All CPU cores | TED video subsystem   | Bus-routed TED character, colour, and screen memory.                                                                                                          |
| 0xF1000-0xF13FF | 1KB   | MMIO                     | VGA registers                           | All CPU cores | VGA subsystem         | Sequencer, CRTC, graphics-controller, attribute-controller, and DAC state.                                                                                    |
| 0xF1400-0xF140F | 16B   | MMIO                     | Host helper                             | All CPU cores | Host-helper subsystem | Security-sensitive command helper gated by runtime configuration.                                                                                             |
| 0xF2000-0xF2017 | 24B   | MMIO                     | ULA registers                           | All CPU cores | ULA subsystem         | Includes paged VRAM data port.                                                                                                                                |
| 0xFA000-0xFBAFF | 6912B | ULA VRAM aperture        | ULA display memory                      | All CPU cores | ULA subsystem         | Separate decoded aperture for ULA VRAM access.                                                                                                                |
| 0xF2100-0xF213F | 64B   | MMIO                     | ANTIC registers                         | All CPU cores | ANTIC subsystem       | ANTIC display-list and DMA control block.                                                                                                                     |
| 0xF2140-0xF21FB | 188B  | MMIO                     | GTIA registers                          | All CPU cores | GTIA subsystem        | GTIA colour, player/missile, priority, and collision state.                                                                                                   |
| 0xF2200-0xF221F | 32B   | MMIO                     | File I/O                                | All CPU cores | File-I/O bridge       | Legacy host-file bridge register block with 32-bit name, data, length, status, error, and read-cap registers.                                                 |
| 0xF22B0-0xF22B7 | 8B    | MMIO64                   | File I/O IE64 extension                 | IE64          | File-I/O bridge       | FILE_DATA_PTR64, a 64-bit data-buffer pointer for IE64 code that reads or writes host files from high guest RAM. The legacy File I/O block remains unchanged. |
| 0xF2220-0xF225F | 64B   | MMIO                     | AROS DOS handler                        | All CPU cores | AROS profile          | AROS DOS integration register block.                                                                                                                          |

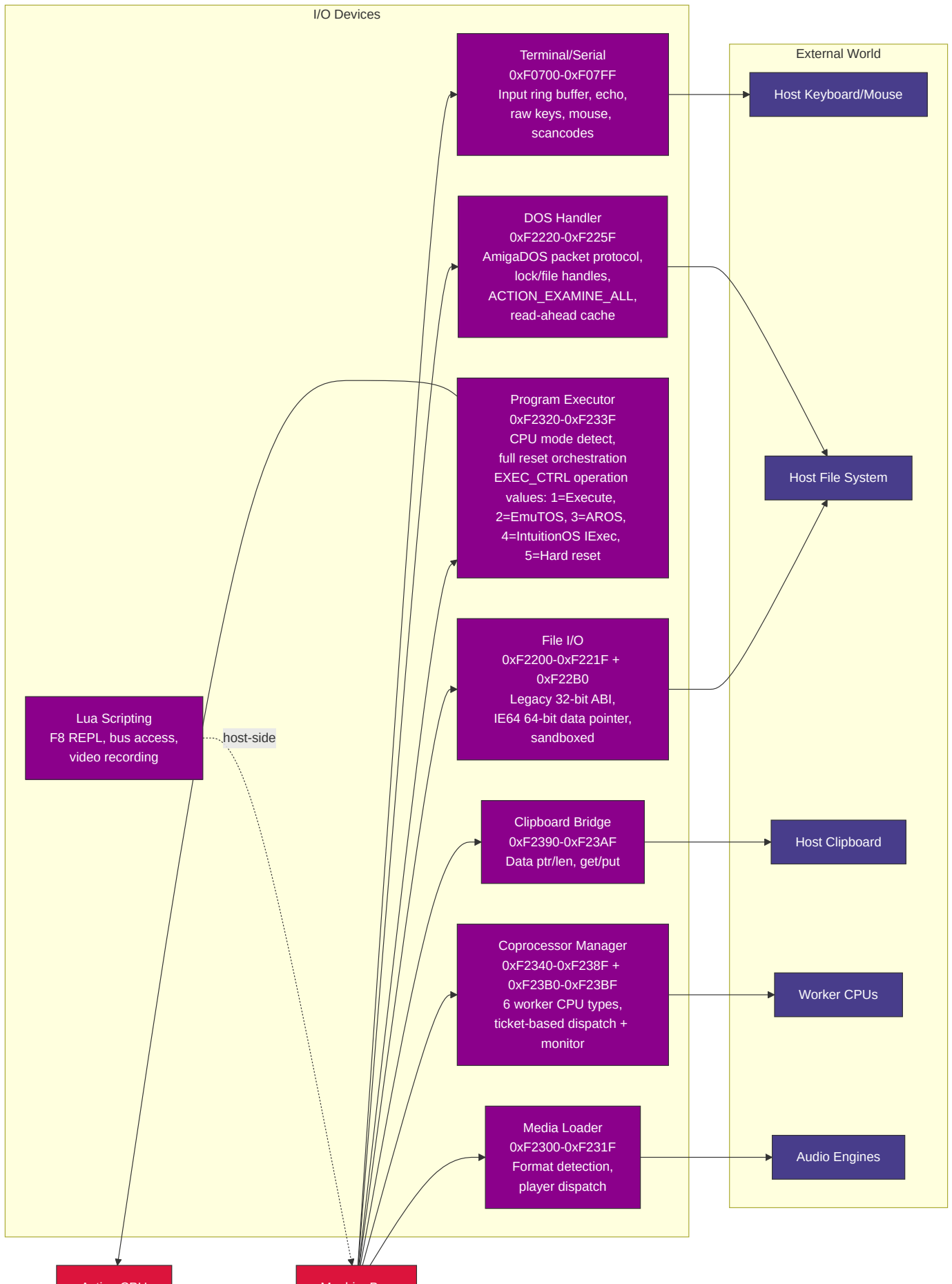
| Range               | Size | Decoded as                      | Architectural use                                  | Visible to                                     | Owner / lifetime                      | Notes                                                                                                                                                         |
|---------------------|------|---------------------------------|----------------------------------------------------|------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xF2260-0xF22AF     | 80B  | MMIO                            | AROS Paula-style DMA shim                          | All CPU cores                                  | AROS profile                          | Audio/DMA compatibility shim.                                                                                                                                 |
| 0xF2300-0xF231F     | 32B  | MMIO                            | Media loader                                       | All CPU cores                                  | Media-loader subsystem                | Format-agnostic media loading control block.                                                                                                                  |
| 0xF2320-0xF233F     | 32B  | MMIO                            | Program executor                                   | All CPU cores                                  | Program executor                      | CPU/program switching control block.                                                                                                                          |
| 0xF2340-0xF238F     | 80B  | MMIO                            | Coprocessor manager                                | All CPU cores                                  | Coprocessor subsystem                 | Worker lifecycle, dispatch, and ticket registers.                                                                                                             |
| 0xF2390-0xF23AF     | 32B  | MMIO                            | Clipboard bridge                                   | All CPU cores                                  | Clipboard bridge                      | Host clipboard integration register block.                                                                                                                    |
| 0xF23B0-0xF23BF     | 16B  | MMIO                            | Coprocessor extended monitor registers             | All CPU cores                                  | Coprocessor subsystem                 | Per-worker monitor/status extension.                                                                                                                          |
| 0xF23C0-0xF23DF     | 32B  | MMIO                            | AROS IRQ diagnostic registers                      | AROS M68K profile while AROS loader is active  | AROS profile                          | Read-only diagnostic MMIO mapped by the AROS loader and unmapped during AROS DMA teardown; not a universal interrupt-controller ABI.                          |
| 0xF23E0-0xF23FF     | 32B  | MMIO                            | Bootstrap HostFS                                   | All CPU cores                                  | Bootstrap profile                     | HostFS boot helper register block.                                                                                                                            |
| 0xF2400-0xF24FF     | 256B | MMIO                            | SYSINFO RAM-size discovery                         | All CPU cores                                  | System information ABI                | Reports total and active visible RAM.                                                                                                                         |
| 0xF2500-0xF257F     | 128B | MMIO                            | AROS host socket bridge                            | AROS M68K profile                              | AROS profile                          | Host-backed <code>bsdsocket.library</code> command bridge.                                                                                                    |
| 0xF2580-0xF259F     | 32B  | MMIO                            | CPU wait service                                   | All CPU cores                                  | Timing/wait ABI                       | CPU wait writes park until the next VBlank edge or a latched <code>RTC_MONO_USEC</code> deadline, capped by a 50 ms safety timeout; reads return 0.           |
| 0xF25A0-0xF25BF     | 32B  | MMIO                            | Coprocessor instance discovery                     | 32-bit-addressing CPU cores and main-CPU BASIC | Coprocessor subsystem                 | Selected type's instance limit, selected-instance state, mailbox layout version, worker window and ring addresses, plus an atomic all-instance liveness mask. |
| 0xF8000-0xF87FF     | 2KB  | MMIO                            | Voodoo 3D registers and palette                    | All CPU cores                                  | Voodoo subsystem                      | 3D control, state, and palette register block.                                                                                                                |
| 0xF8140-0xF823F     | 256B | MMIO                            | Voodoo fog table                                   | All CPU cores                                  | Voodoo subsystem                      | 64 entries x 4 bytes.                                                                                                                                         |
| 0x100000-0x5FFFFFFF | 5MB  | Shared RAM / VRAM-backed region | Main video framebuffer and graphics-visible memory | All CPU cores                                  | Video subsystem plus guest convention | Subranges may be reserved for coprocessor worker buffers.                                                                                                     |

| Range                  | Size  | Decoded as                        | Architectural use                | Visible to    | Owner / lifetime               | Notes                                                                                                         |
|------------------------|-------|-----------------------------------|----------------------------------|---------------|--------------------------------|---------------------------------------------------------------------------------------------------------------|
| 0x200000 - 0x27FFFF    | 512KB | Shared RAM                        | IE32 worker area                 | All CPU cores | Coprocessor convention         | Lies inside the graphics-visible shared-memory range; not a separate address space.                           |
| 0x280000 - 0x2FFFFFF   | 512KB | Shared RAM                        | M68K worker area                 | All CPU cores | Coprocessor convention         | Lies inside the graphics-visible shared-memory range; not a separate address space.                           |
| 0x300000 - 0x30FFFF    | 64KB  | Shared RAM                        | 6502 worker area                 | All CPU cores | Coprocessor convention         | Lies inside the graphics-visible shared-memory range; not a separate address space.                           |
| 0x310000 - 0x31FFFF    | 64KB  | Shared RAM                        | Z80 worker area                  | All CPU cores | Coprocessor convention         | Lies inside the graphics-visible shared-memory range; not a separate address space.                           |
| 0x320000 - 0x39FFFF    | 512KB | Shared RAM                        | x86 worker area                  | All CPU cores | Coprocessor convention         | Lies inside the graphics-visible shared-memory range; not a separate address space.                           |
| 0x3A0000 - 0x41FFFF    | 512KB | Shared RAM                        | IE64 worker area                 | All CPU cores | Coprocessor convention         | Lies inside the graphics-visible shared-memory range; not a separate address space.                           |
| 0x420000 - 0x49FFFF    | 512KB | Shared RAM                        | M68K worker instance 1 area      | All CPU cores | Coprocessor convention         | Second M68K worker RAM; ordinary big-endian worker memory inside the graphics-visible shared-memory range.    |
| 0x4A0000 - 0x51FFFF    | 512KB | Shared RAM                        | x86 worker instance 1 area       | All CPU cores | Coprocessor convention         | Second x86 worker RAM inside the graphics-visible shared-memory range.                                        |
| 0x520000 - 0x59FFFF    | 512KB | Shared RAM                        | IE64 worker instance 1 area      | All CPU cores | Coprocessor convention         | Second IE64 worker RAM inside the graphics-visible shared-memory range.                                       |
| 0x790000 - 0x792FFF    | 12KB  | Shared RAM                        | Coprocessor mailbox ring buffers | All CPU cores | Coprocessor subsystem          | Twelve ring slots at a 0x400 stride; shared request/completion rings outside the main graphics-visible range. |
| 0x800000 - 0x80FFFF    | 64KB  | Shared RAM                        | Media-loader staging buffer      | All CPU cores | Media-loader subsystem         | Reservation at the base of the AROS fast-memory range when that profile is active.                            |
| 0x800000 - 0x1DFFFFFF  | 22MB  | Shared RAM                        | AROS fast memory                 | All CPU cores | AROS profile                   | Profile-owned allocation range, not a distinct per-core map.                                                  |
| 0x1E00000 - 0x5DFFFFFF | 64MB  | Shared RAM / video-profile memory | AROS video RAM                   | All CPU cores | AROS profile / video subsystem | Profile-owned video allocation range within shared physical memory.                                           |

## 7. I/O Peripherals

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The File I/O ABI keeps the original 32-bit register block at 0xF2200-0xF221F for every CPU. IE64 adds FILE\_DATA\_PTR64 at 0xF22B0 as a separate 64-bit MMIO register for data buffers in high guest RAM, including buffers allocated from the native AOT arena. Non-IE64 software does not need to know about the extension.

## Coprocessor Worker Dispatch

The coprocessor manager supports 6 worker CPU types (IE32, IE64, 6502, M68K, Z80, x86) with ticket-based job dispatch and mailbox ring buffers at 0x790000. Each worker type has a dedicated shared-memory reservation in the unified physical map (see memory map above), not a private per-core address space. The main CPU enqueues work via MMIO writes; workers execute independently and post results back through their mailbox slots. Each ring has 16 descriptor slots but uses one slot to distinguish full from empty, so it can hold 15 queued requests at once. When COPROC\_IRQ\_CTRL bit 0 is set and an M68K IRQ target plus completion watcher have been configured, the coprocessor asserts M68K interrupt level 6 on job completion, with the finished ticket ID readable from COPROC\_COMPLETED\_TICKET. Non-M68K modes should observe completion through POLL or WAIT.

COPROC\_INSTANCE selects a worker instance together with COPROC\_CPU\_TYPE. The JIT-capable worker types (M68K, x86, IE64) support instances 0 and 1; IE32, 6502, and Z80 support instance 0 only. Instance-1 windows are M68K 0x420000-0x49FFFF, x86 0x4A0000-0x51FFFF, and IE64 0x520000-0x59FFFF, each 512 KiB. The mailbox holds twelve ring slots at a uniform 0x400 stride, and the ring index is  $\text{cpuTypeIndex} * 2 + \text{instance}$  with no special case (the 0x400 stride exceeds a ring's 0x308 content, so no ring's last slot overflows into its neighbour). COPROC\_SELECTED\_STATE reports whether the selected worker is running, while COPROC\_INSTANCE\_STATE reports all live instances atomically without changing COPROC\_CPU\_TYPE or COPROC\_INSTANCE; the aggregate COPROC\_WORKER\_STATE sets a per-type bit when any instance of that type is running. Start, stop, enqueue, ring depth, uptime, and ticket ownership are scoped to the selected instance. A worker must echo the mailbox layout version (COPROC\_MAILBOX\_VERSION) into its ring header at startup or the manager refuses to route it work and START fails with COPROC\_ERR\_STALE\_WORKER.

The coprocessor discovery block at 0xF25A0-0xF25BF reports the selected type's instance limit, selected-instance state, mailbox layout version, worker window, worker ring, and an atomic all-instance liveness mask. The mailbox contains twelve 0x400-byte ring slots from 0x790000 through 0x792FFF; each ring publishes layout version 1 and a worker must acknowledge that version before START succeeds.

Instance-1 windows are M68K 0x420000-0x49FFFF, x86 0x4A0000-0x51FFFF, and IE64 0x520000-0x59FFFF, each 512 KiB; the ring index is  $\text{cpuTypeIndex} * 2 + \text{instance}$ , COPROC\_SELECTED\_STATE reports the selected worker, and COPROC\_INSTANCE\_STATE reports all live instances without changing the selectors.

M68K workers initialise their own stack bounds and supervisor/user stack pointers and arm the M68K JIT. Coprocessor shared-window pages route through memory helpers so their byte-swap rules match interpreted execution; both M68K worker RAM windows are excluded from that shared-window treatment and remain ordinary big-endian worker RAM. M68K block compilation is serialised across the main CPU and worker instances.

## Lua Scripting

The Lua scripting engine (script\_engine.go) runs in its own goroutine and provides host-side automation over the shared 32-bit bus/MMIO surface plus CPU-adapter debugger APIs. Its mem.\* helpers are raw 32-bit bus helpers, not an above-4GiB IE64 RAM or CPU-virtual-address API. It supports an F8 REPL for interactive debugging, video recording, and direct chip register manipulation. Scripts use the .ies extension and are loaded via the IE Script Engine.

## Recording Pipeline

Video recording sends compositor frames and 44.1 kHz mono audio to FFmpeg on separate pipes. The audio tap observes the mixed output without advancing the sound engines a second time. Recording follows wall-clock time; after an encoder stall the recorder discards missed video-frame debt and matching oldest buffered audio while preserving the newest output batch. This prevents a blocked encoder from replaying stale media in an unbounded catch-up burst after it resumes. Video and audio recording pumps are independent; frozen or unchanged video is held, and audio starvation beyond 500 ms produces silence instead of stalling.

## IEMon Reverse-Debug Snapshot Contract

IEMon whole-machine snapshots enumerate the monitor CPU registry by stable CPU id and label, so profiles with omitted CPUs or multiple coprocessors restore by identity rather than by CPU type. Shared bus RAM and IE64 backing memory are stored as sparse 4 KiB pages. The reverse-history chain keeps full sparse checkpoints plus page deltas anchored to retained checkpoints; the monitor materialises deltas before `rg` or `rt` applies a restore.

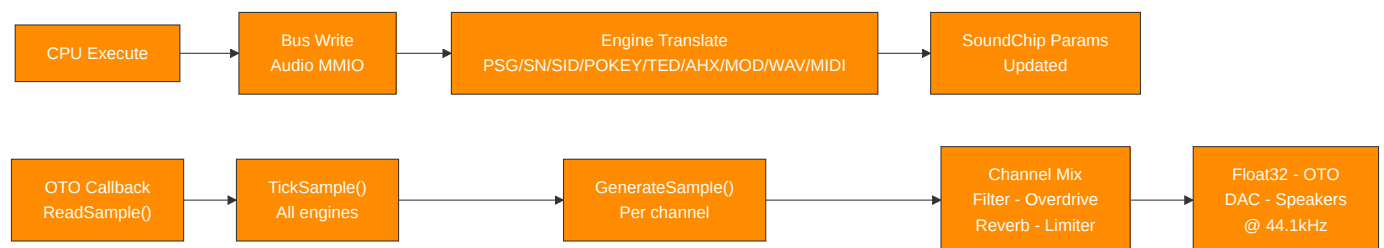
Mutable devices join the snapshot contract through `MachineMonitor.RegisterSnapshotDevice`. Each device supplies a stable name, a version, and an opaque guest-visible state blob. Restore fails closed if a captured device is absent, preventing partial reverse-debug restores. The production monitor registers the main video chip, sound chip, terminal MMIO, command-style host helpers, compatibility audio/video engines, and AROS clipboard/audio-DMA bridges when present; new guest-visible timers, DMA engines, IRQ controllers, and host bridges must register their own versioned blobs before they are considered covered by whole-machine reverse debugging.

## 8. Data Flow

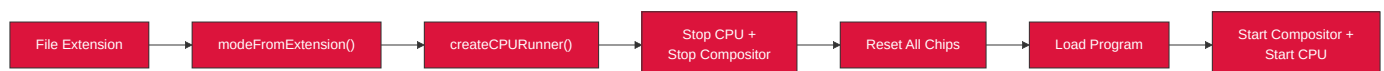
### Video Pipeline



### Audio Pipeline



### CPU Mode Switching



## 9. Concurrency Model and System Timing

### Goroutine Inventory

| Goroutine           | Clock Source                                      | Rate                           | Synchronisation                                                                      |
|---------------------|---------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------|
| CPU execution       | Free-running for <code>running.Load()</code> loop | As fast as Go scheduler allows | MMIO writes invoke I/O callbacks under per-chip <code>mu</code>                      |
| VideoChip refresh   | <code>time.NewTicker</code>                       | 60Hz                           | VBlank timing, copper advancement, and blitter IRQ latency                           |
| VGA render          | <code>time.NewTicker</code>                       | 60Hz                           | Renders into triple-buffer slot, publishes via <code>sharedIdx.Swap()</code>         |
| ULA render          | <code>time.NewTicker</code>                       | 60Hz                           | Same triple-buffer pattern                                                           |
| TED render          | <code>time.NewTicker</code>                       | 60Hz                           | Same triple-buffer pattern                                                           |
| ANTIC render        | <code>time.NewTicker</code>                       | 60Hz                           | Same triple-buffer pattern                                                           |
| Compositor          | <code>time.NewTicker</code>                       | 60Hz                           | Calls <code>GetFrame()</code> (atomic swap) on each source, blends, sends to display |
| OTO audio thread    | Host audio hardware callback                      | 44.1kHz                        | Calls <code>SoundChip.ReadSample()</code> -> <code>GenerateSample()</code> directly  |
| Terminal output     | <code>time.NewTicker</code>                       | 100Hz                          | Flushes output buffer to host terminal                                               |
| Coprocessor workers | On-demand                                         | Varies                         | Independent CPUs with mailbox ring buffers                                           |
| Script engine       | Event-driven                                      | Varies                         | Lua goroutine with frame-sync channel                                                |

### Three Independent Clocks

CPU: Free-running (no fixed clock -- instruction throughput varies with host speed)  
Video: 6 video sources plus compositor tick at 60Hz; non-VideoChip sources use lock-free triple-buffer handoff  
Audio: OTO hardware callback drives sample generation at 44.1kHz -- no IE-owned goroutine

### Synchronisation Model

**Hot paths** (lock-free): - `atomic.Bool` - CPU running, chip enabled, `compositorManaged` - `atomic.Int32` - triple-buffer  
`sharedIdx` - `atomic.Pointer` - selected CPU pointer

**Cold paths** (mutex-protected): - `sync.Mutex` (`chip.mu`, `video.mu`) - configuration changes, setup, stop

**CPU-to-Video:** - CPU writes to VRAM/MMIO invoke I/O callbacks under per-chip `mu` - Video chips read VRAM lock-free (snapshot-render pattern: copy under lock, render without lock)

**CPU-to-Audio:** - CPU writes to audio MMIO update channel parameters under `chip.mu` - OTO callback reads parameters atomically or under `chip.mu` for `GenerateSample()`

**No CPU-video vsync coupling:** - CPU runs ahead freely - no wait-for-vblank blocking - `WAIT` register polls `vblankActive` `atomic.Bool` without blocking the CPU

## Appendix: Key Source Files

| File                                                | Role                                                                                                                                             |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| registers.go                                        | Master I/O address map - all region boundaries                                                                                                   |
| machine_bus.go                                      | MachineBus: autodetected guest RAM, MapIO, ioPageBitmap, Read/Write, SYSINFO accessors                                                           |
| memory_sizing.go                                    | Boot-time guest RAM autodetection: total guest RAM + active visible RAM with platform reserves                                                   |
| profile_bounds.go                                   | Source-owned profile bounds for EmuTOS, AROS, IE64 BASIC                                                                                         |
| emulator_cpu.go                                     | EmulatorCPU interface definition                                                                                                                 |
| video_interface.go                                  | VideoSource, VideoOutput, ScanlineAware interfaces                                                                                               |
| video_compositor.go                                 | Compositor pipeline, Z-order blending                                                                                                            |
| video_chip.go                                       | VideoChip + Copper + Blitter + Palette                                                                                                           |
| video_vga.go                                        | VGA engine (Sequencer, CRTC, GC, AC, DAC)                                                                                                        |
| video_ula.go                                        | ULA engine (Spectrum display)                                                                                                                    |
| video_ted.go                                        | TED video engine                                                                                                                                 |
| video_antic.go                                      | ANTIC + GTIA engines                                                                                                                             |
| video_voodoo.go                                     | Voodoo 3D pipeline                                                                                                                               |
| audio_chip.go                                       | SoundChip (10-channel synthesis)                                                                                                                 |
| psg_engine.go                                       | PSG / AY-3-8910                                                                                                                                  |
| sn76489_chip.go                                     | SN76489 programmable sound generator                                                                                                             |
| sid_engine.go                                       | SID 6581/8580                                                                                                                                    |
| pokey_engine.go                                     | POKEY                                                                                                                                            |
| ted_engine.go                                       | TED audio                                                                                                                                        |
| ahx_engine.go                                       | AHX replayer                                                                                                                                     |
| mod_player.go                                       | MOD player                                                                                                                                       |
| wav_player.go                                       | WAV player                                                                                                                                       |
| midi_player.go / midi_engine.go /<br>midi_parser.go | SMF .mid/.midi and Doom .mus; fixed RawlandMini IE SoundChip GM-style/chiptune interpretation, 10 active MIDI voices with deterministic stealing |
| midi_live.go                                        | Generic live-MIDI MMIO port: running-status byte-stream parser feeding the shared MIDIEngine (all cores + BASIC)                                 |
| atari_midi_acia.go                                  | EmuTOS-only MC6850 MIDI ACIA shim (\$FFFC04/06, low-16 + 24-bit aliases) bridging output-only MIDI bytes into the shared LiveMIDI/MIDIEngine     |
| cpu_ie32.go                                         | IE32 CPU                                                                                                                                         |
| cpu_ie64.go                                         | IE64 CPU + interpreter                                                                                                                           |
| fpu_ie64.go                                         | IE64 FPU (16 x float32)                                                                                                                          |
| jit_emit_arm64.go                                   | IE64 JIT ARM64 emitter                                                                                                                           |

| File                   | Role                                                                       |
|------------------------|----------------------------------------------------------------------------|
| jit_emit_amd64.go      | IE64 JIT x86-64 emitter                                                    |
| cpu_m68k.go            | M68K 68020                                                                 |
| fpu_m68881.go          | M68881 FPU (8 x float64)                                                   |
| cpu_z80.go             | Z80                                                                        |
| cpu_z80_runner.go      | Z80 bus adapter, port I/O bridge, bank windows                             |
| cpu_six5go2.go         | 6502 + bus adapter, ioTable dispatch, bank windows                         |
| cpu_x86.go             | x86                                                                        |
| cpu_x86_runner.go      | x86 bus adapter, shared port I/O bridge, bank windows, and VGA VRAM window |
| fpu_x87.go             | x87 FPU (8 x float64 stack)                                                |
| terminal_io.go         | Terminal / Serial / Mouse / Keyboard                                       |
| file_io.go             | File I/O                                                                   |
| program_executor.go    | Program Executor                                                           |
| coprocessor_manager.go | Coprocessor Manager                                                        |
| clipboard_bridge.go    | Clipboard Bridge                                                           |
| media_loader.go        | Media format detection + player dispatch                                   |
| script_engine.go       | Lua scripting engine                                                       |
| aros_loader.go         | AROS ROM boot manager                                                      |
| aros_dos_intercept.go  | AmigaDOS packet handler (MMIO)                                             |
| aros_audio_dma.go      | AROS Paula-style DMA shim                                                  |